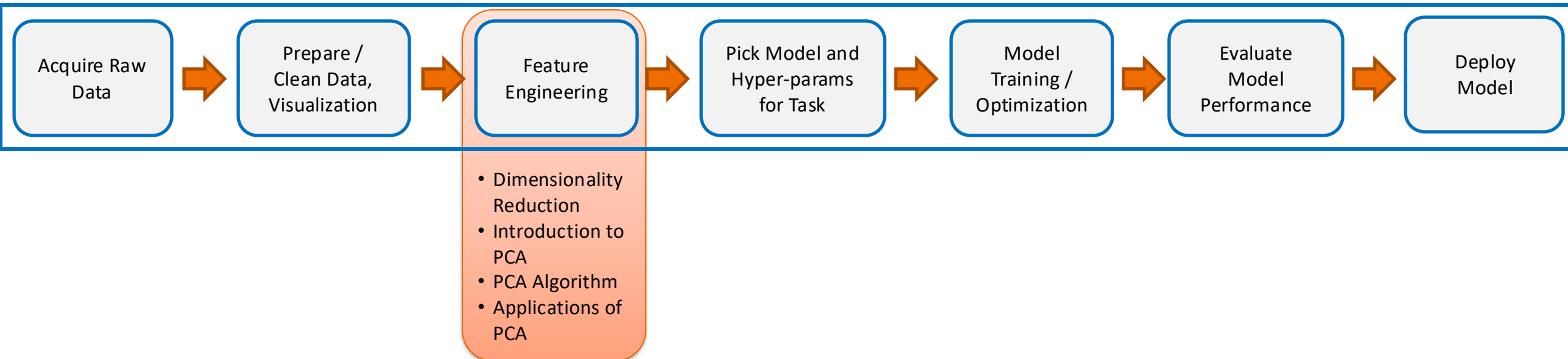
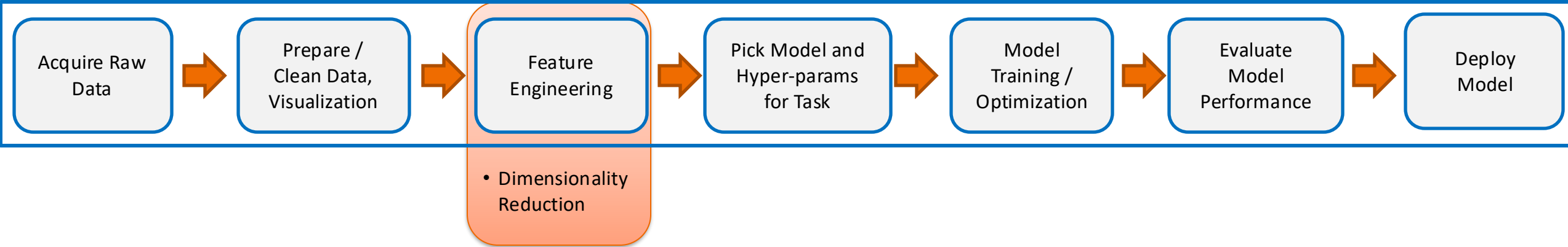


Focus for this lecture

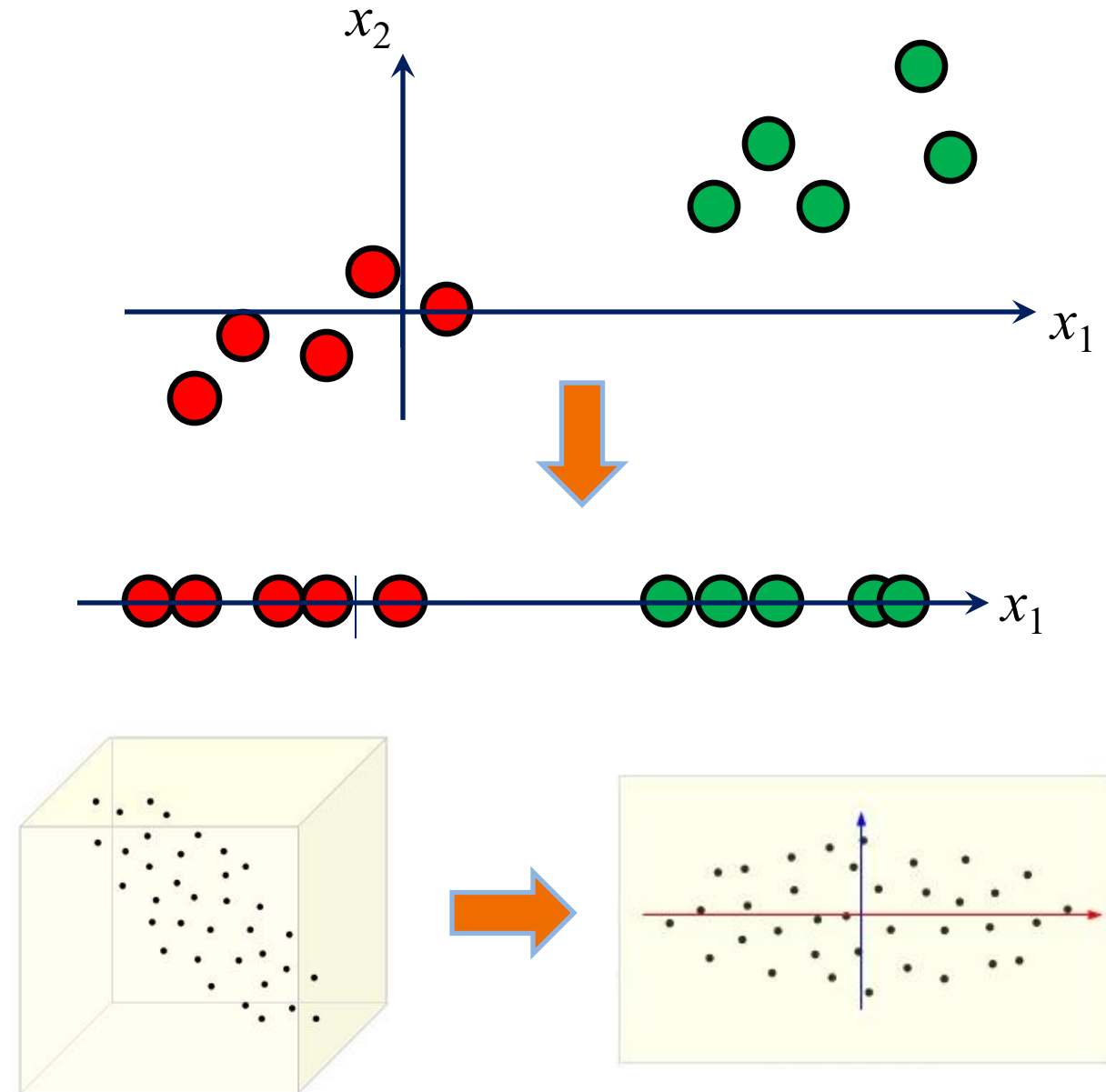




Dimensionality Reduction

Reducing Dimensions

- **Feature Selection:**
 - Choose the "best" features from your data
- **Feature Extraction:**
 - Initial set of measured data and builds derived features intended to be **informative** and **non-redundant**
- **Feature Visualization:**
 - How are the 'best' features distributed in 1D/2D/3D ?



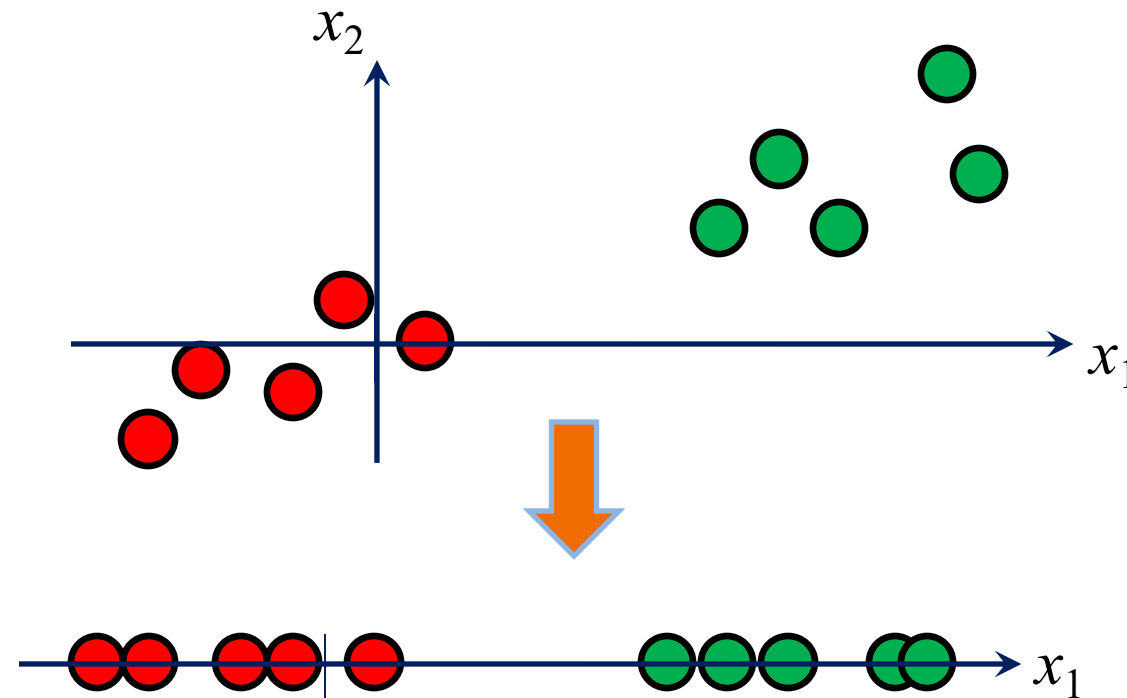
Selecting and Extracting Features

$$\begin{bmatrix} x_1 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

Selecting first and third feature

$$\begin{bmatrix} x_1 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

Selecting first and fourth feature

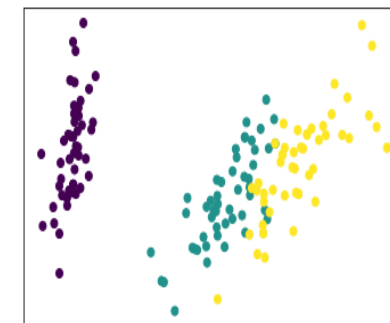


NOTE: Data samples are color-coded by their class label. But label info is not used for feature selection.

Selecting and Extracting Features

$$\begin{bmatrix} x_1 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

Selecting first and third feature



$$\begin{bmatrix} x_1 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

Selecting first and fourth feature

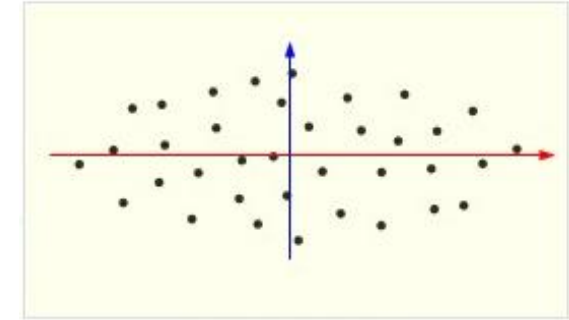
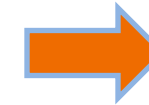
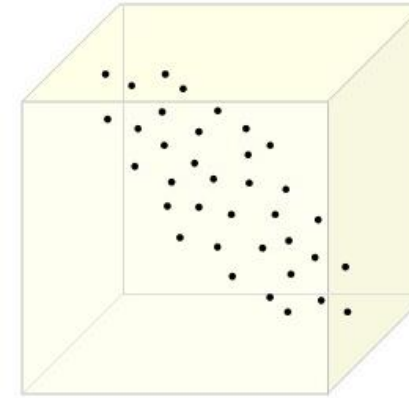
NOTE: Data samples are color-coded by their class label. But label info is not used for feature selection.

Selecting and Extracting Features

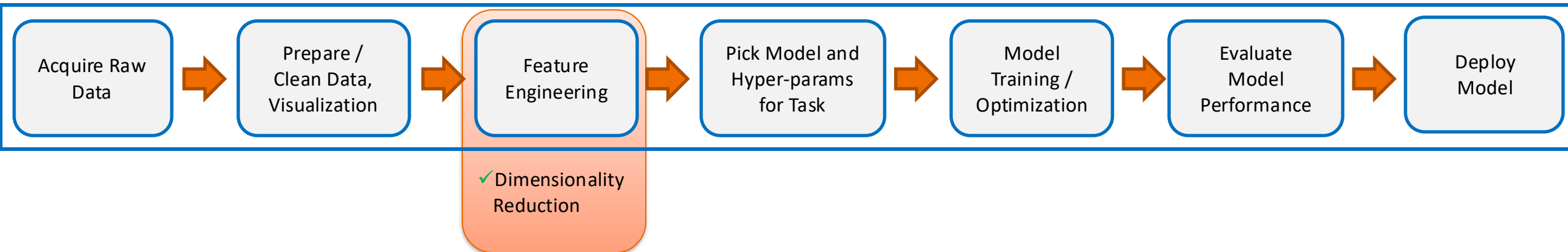
$$\begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = \begin{bmatrix} 0.1 & 0.2 & 0.3 \\ 0.0 & 0.4 & 0.2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

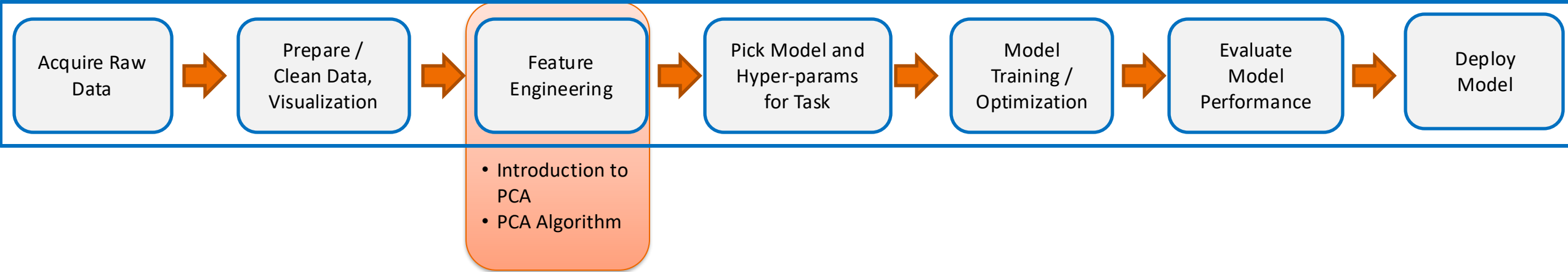
New Features as linear combination of old Features

$$X' = AX$$



Journey so far...





Feature Extraction

Introduction to Principal Component Analysis (PCA)

PCA: A Toy Example

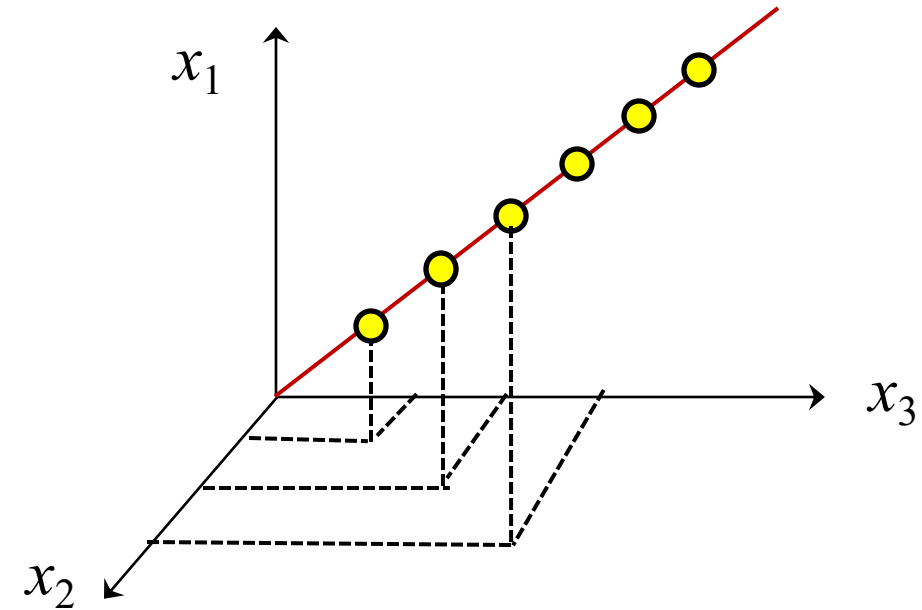
- Consider the following dataset:

1	4	3	6	5	2
2	8	6	12	10	4
3	12	9	18	15	6

PCA: A Toy Example

- All these points fall on a line: a 1-dimensional subspace of the original 3D space:

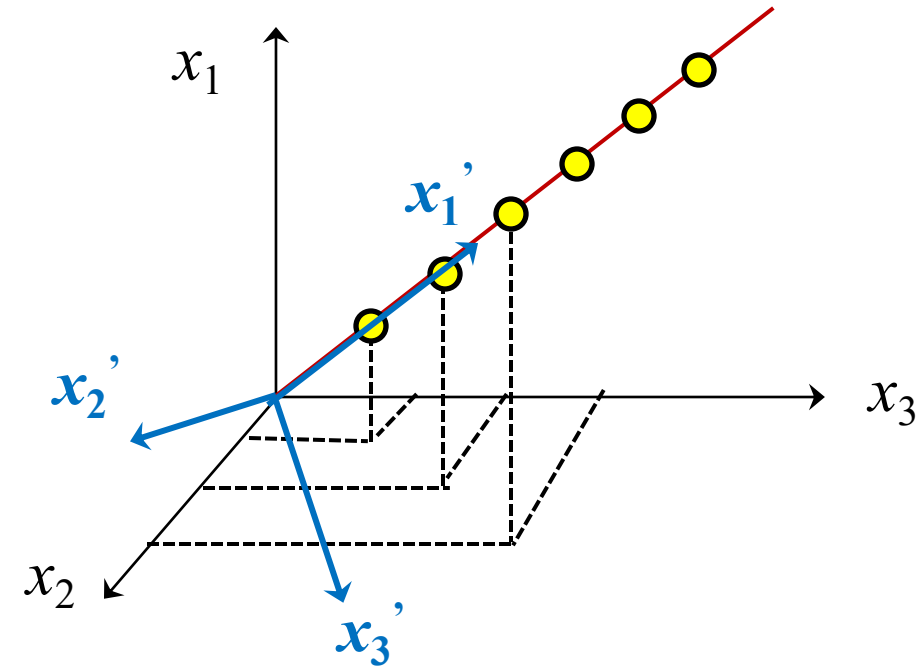
1	2	3	4	5	6
2	4	6	8	10	12
3	6	9	12	15	18



PCA: A Toy Example

- Consider a new co-ordinate system with one axis along the line
- All co-ordinates except the first one are zeros now.

3.7	7.5	11.2	15	18.7	22.4
0	0	0	0	0	0
0	0	0	0	0	0



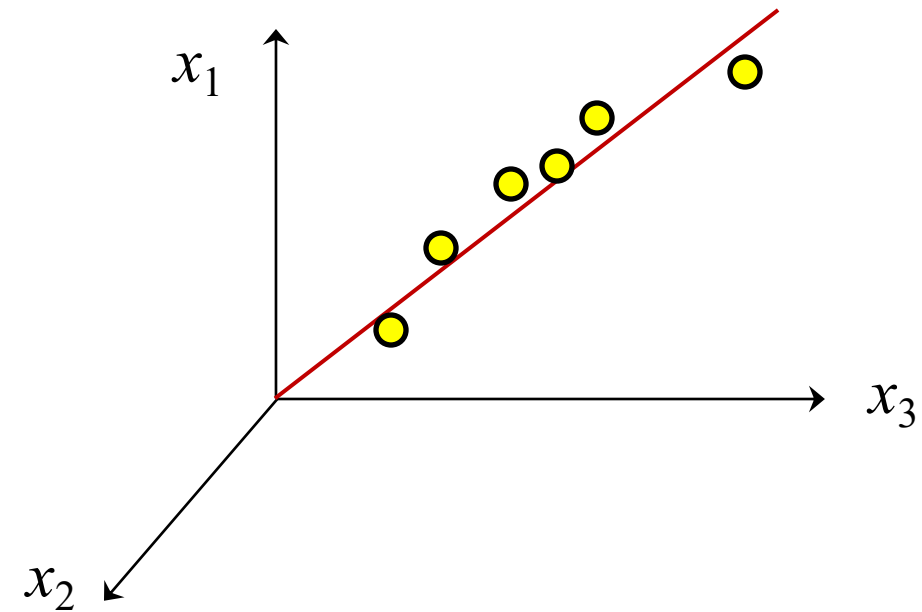
PCA: Toy Example - 2

- Consider the following dataset:

1	4	3	5.7	5.1	2.2
2	7.9	5.8	12	9.9	4.1
3.1	12	9	18	15	6.3

PCA: Toy Example - 2

1	4	3	5.7	5.1	2.2
2	7.9	5.8	12	9.9	4.1
3.1	12	9	18	15	6.3

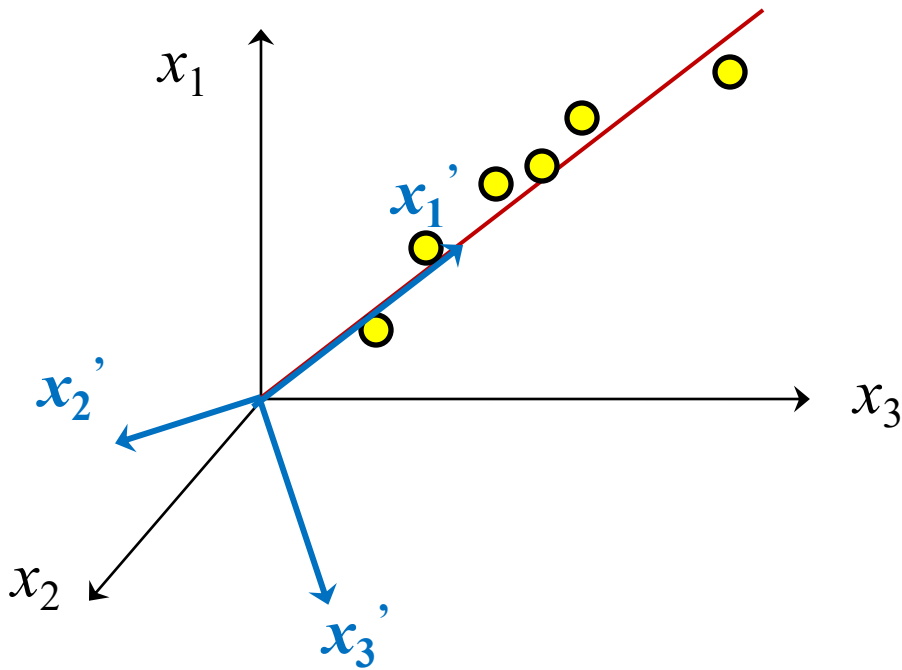


PCA: Toy Example - 2

1	4	3	5.7	5.1	2.2
2	7.9	5.8	12	9.9	4.1
3.1	12	9	18	15	6.3

3.61	7.4	11.1	15.0	18.4	22.4
0.2	0.4	0.9	0.7	0.8	0.3
0.1	0.1	0.1	0.1	0.1	0.1

NOTE: These values are made up. Not exact.



Variance

Data values x	Mean $\bar{x}$	$x - \bar{x}$	$(x - \bar{x})^2$
7	16	-9	81
11	16	-5	25
11	16	-5	25
15	16	-1	1
20	16	4	16
20	16	4	16
28	16	12	144

Variance: s^2

$$s^2 = \frac{\sum (x - \bar{x})^2}{n - 1} = \frac{308}{7 - 1} = \frac{308}{6} =$$

Sample Variance:

$$s^2 = \frac{\sum (x - \bar{x})^2}{n - 1}$$

Standard Deviation:

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

n = Sample size

$$n = 7$$

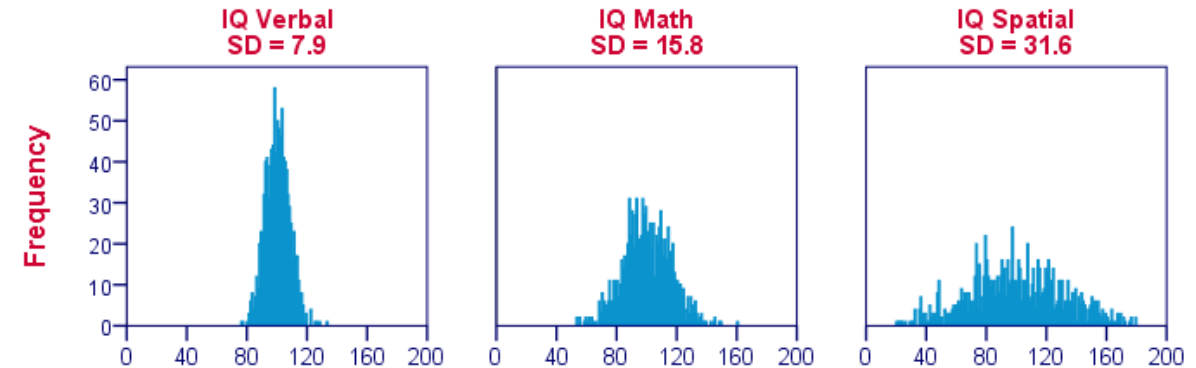
$$\text{Mean} = \frac{\sum x}{n}$$

$$\bar{x} = 16$$

Mean = 'Average' value

S.D = Average deviation of samples from mean

Histograms for IQ Test Components



Covariance : m samples, n features

Vectors 1 and 3

Cell (3, 1) or (1, 3)

$$A = \begin{bmatrix} 1 & 3 & 5 \\ 5 & 4 & 1 \\ 3 & 8 & 6 \end{bmatrix}$$

Covariance

$$\begin{bmatrix} 2.67 & 0.67 & -2.67 \\ 0.67 & 4.67 & 2.33 \\ -2.67 & 2.33 & 4.67 \end{bmatrix}$$

Covariance Matrix

$$\begin{bmatrix} 0.39701 & 0.51117 \\ 0.55582 & 0.93003 \\ 0.59403 & 0.96645 \\ 0.51544 & 0.29759 \\ 0.85313 & 0.18118 \\ 0.88564 & 0.69114 \end{bmatrix}$$

$$\begin{pmatrix} & M1 & M2 & M3 & \dots & Mn \\ S1 & q_{1,1} & q_{1,2} & q_{1,3} & \dots & q_{1,n} \\ S2 & q_{2,1} & q_{2,2} & q_{2,3} & \dots & q_{2,n} \\ S3 & q_{3,1} & q_{3,2} & q_{3,3} & \dots & q_{3,n} \\ \dots & \dots & \dots & \dots & \dots & \dots \\ Sm & q_{m,1} & q_{m,2} & q_{m,3} & \dots & q_{m,n} \end{pmatrix}$$



$$C = \begin{pmatrix} \text{cov}(M_1, M_1) & \text{cov}(M_1, M_2) & \dots & \text{cov}(M_1, M_n) \\ \text{cov}(M_2, M_1) & \text{cov}(M_2, M_2) & \dots & \text{cov}(M_2, M_n) \\ \vdots & \vdots & \ddots & \vdots \\ \text{cov}(M_n, M_1) & \text{cov}(M_n, M_2) & \dots & \text{cov}(M_n, M_n) \end{pmatrix}_{n \times n}$$

N-dimensional Covariance Matrix

Variance:

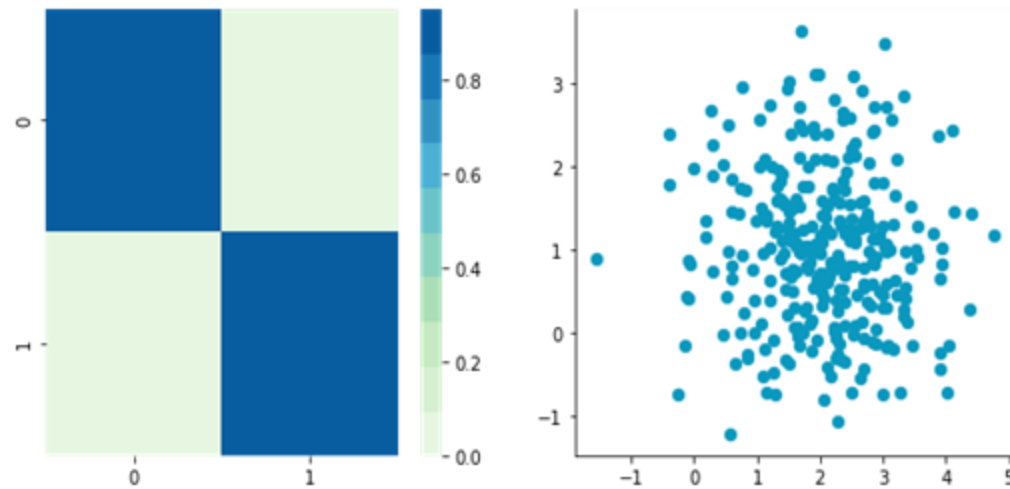
$$s^2 = \frac{\sum (\bar{X} - X_i)^2}{N}$$

Covariance:

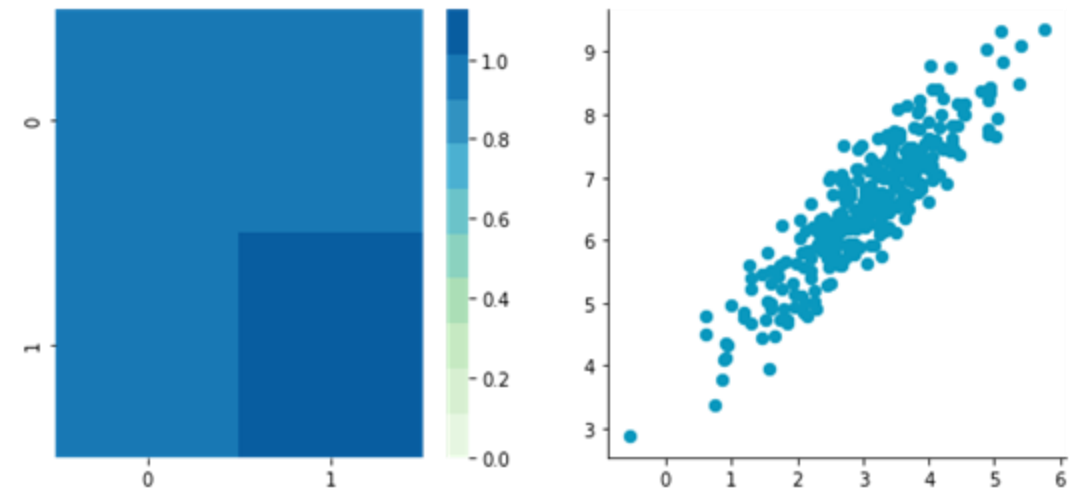
$$\text{cov}(X, Y) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

$$\text{Cov}(M_a, M_b) = \frac{1}{m} \sum_{i=1}^m (q_{i,a} - \bar{q}_a)(q_{i,b} - \bar{q}_b)$$

Covariance Matrix



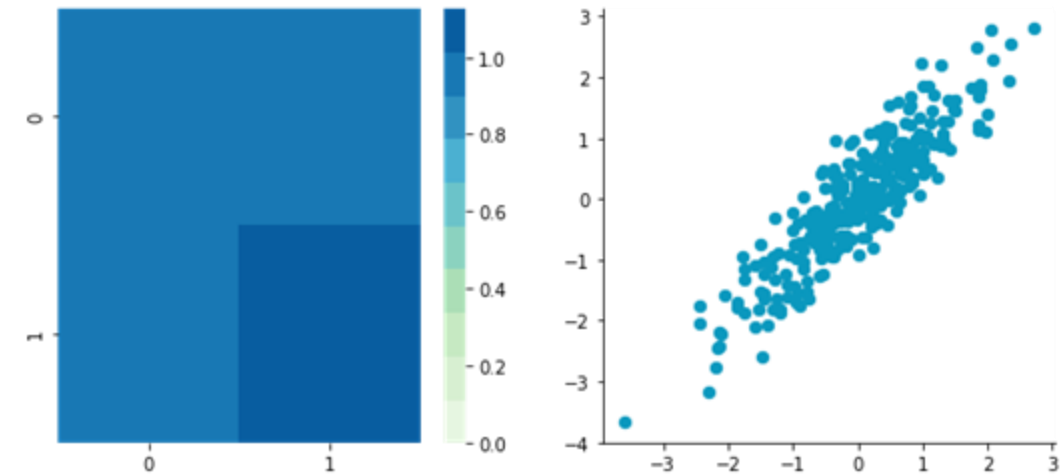
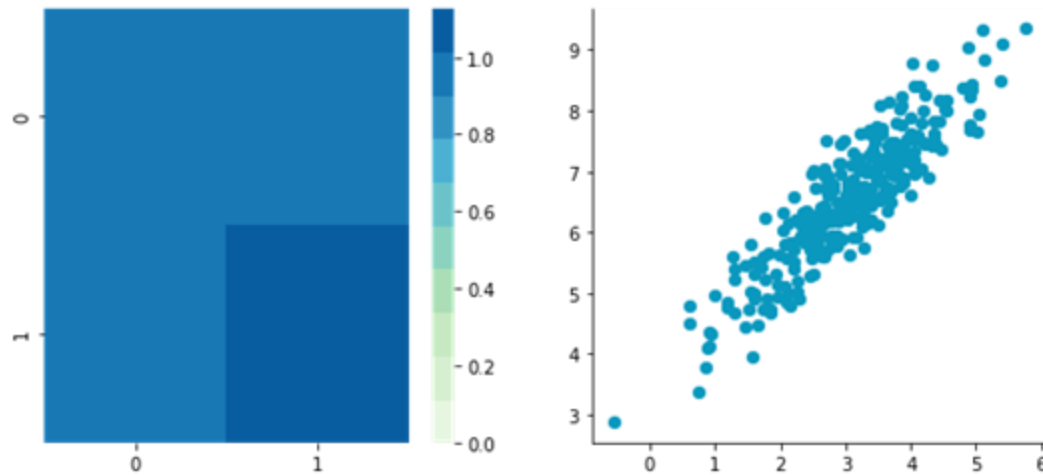
$$C = \begin{bmatrix} +0.95 & -0.04 \\ -0.04 & +0.87 \end{bmatrix}$$



$$C = \begin{bmatrix} +0.95 & +0.92 \\ +0.92 & +1.12 \end{bmatrix}$$

Mean Normalization

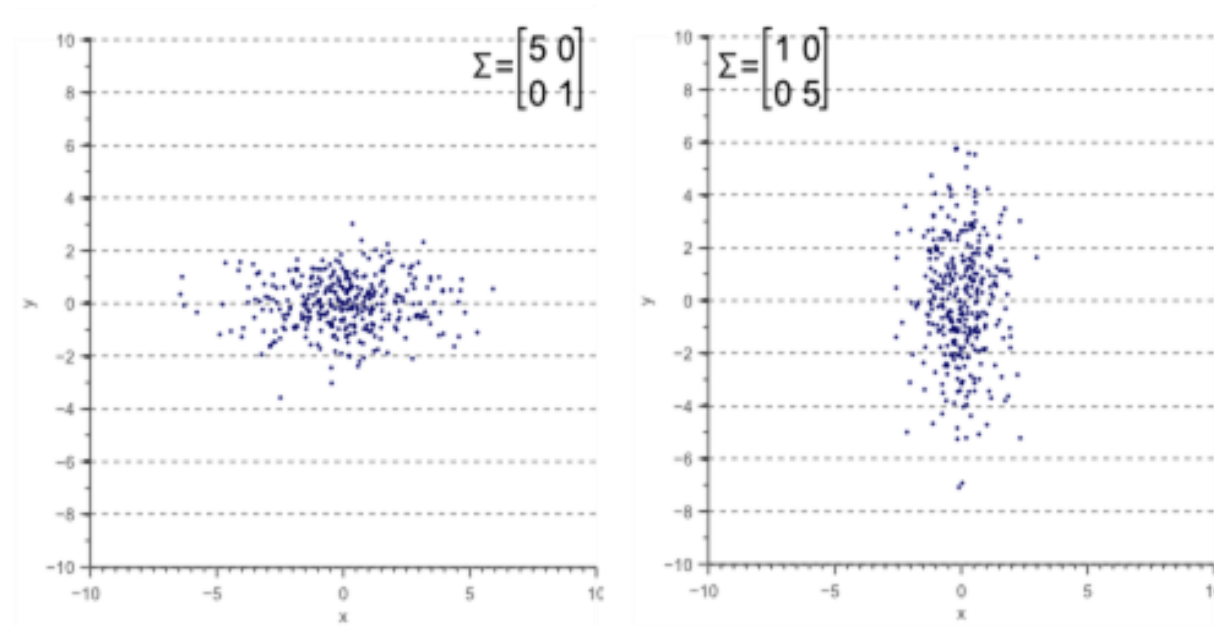
$$\mathbf{X}' = \mathbf{X} - \bar{x}$$



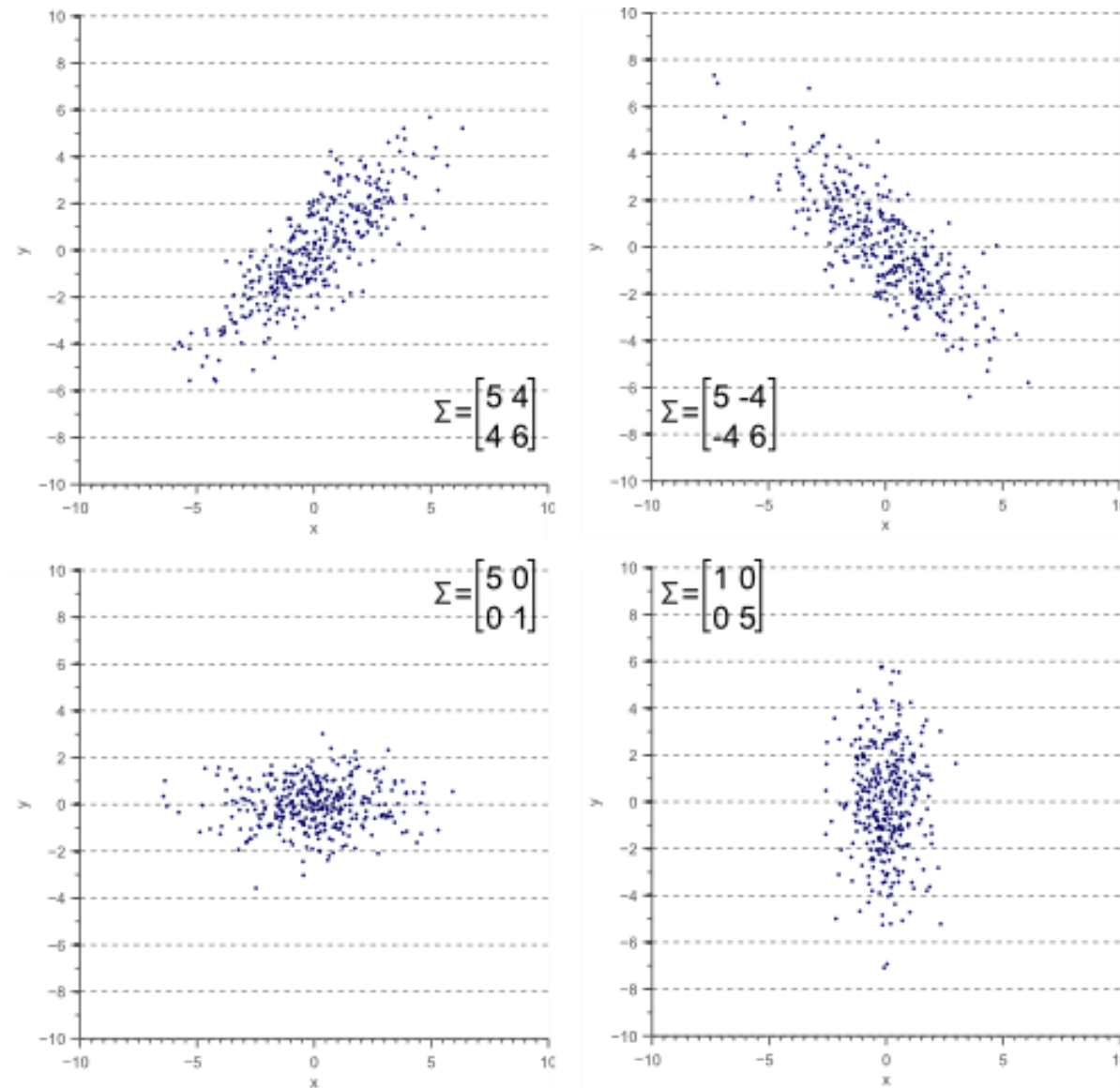
$$C = \begin{bmatrix} +0.95 & +0.92 \\ +0.92 & +1.12 \end{bmatrix}$$

$$C = \begin{bmatrix} +0.95 & +0.92 \\ +0.92 & +1.12 \end{bmatrix}$$

Covariance Matrix encodes spread and orientation data

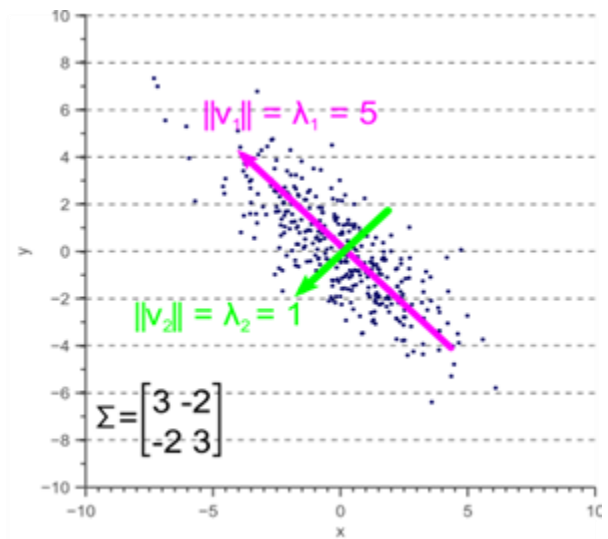
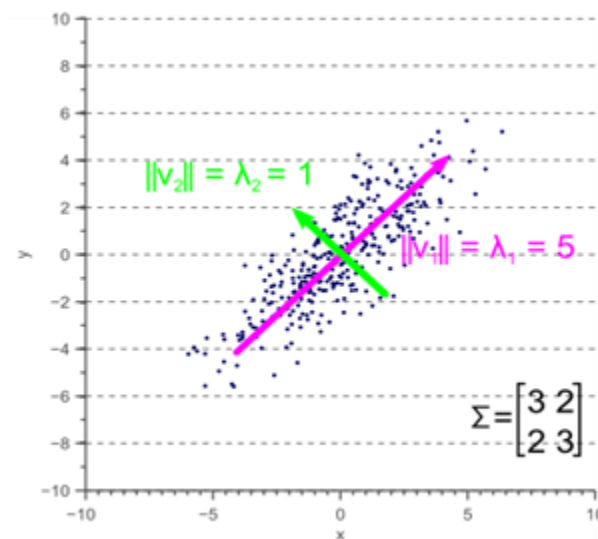
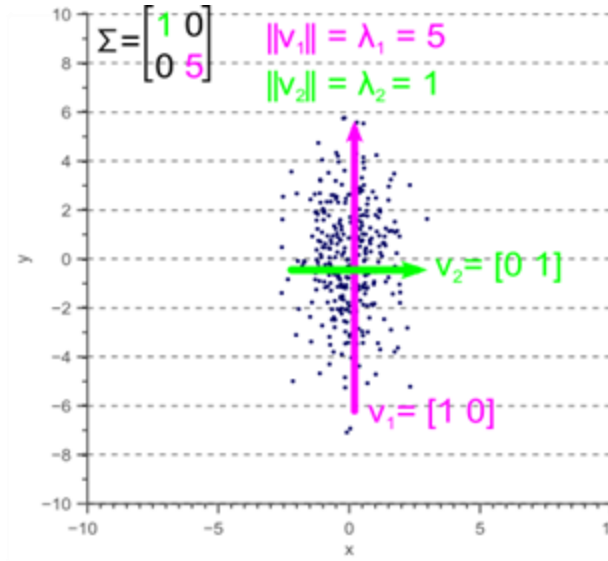
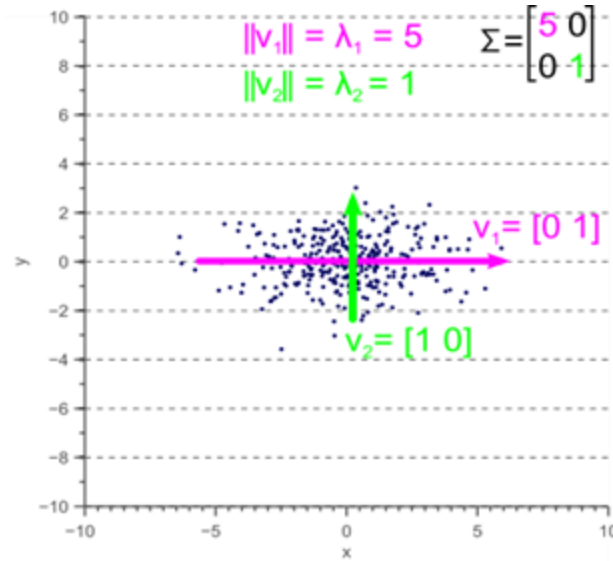


Covariance Matrix encodes spread and orientation data



Eigen-analysis of Covariance Matrix

v_1, v_2 : Principal Components



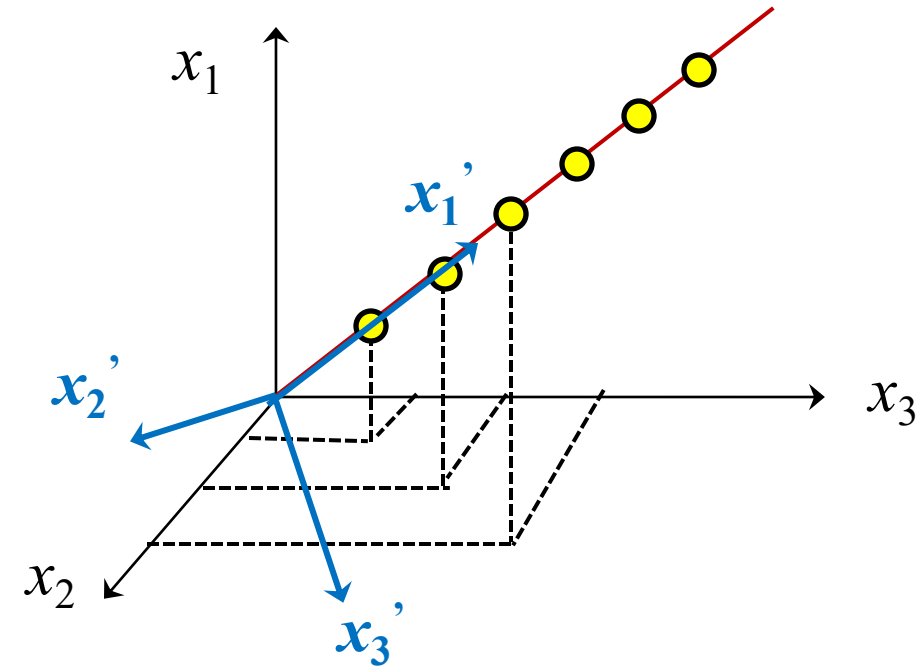
$$\Sigma \vec{v} = \lambda \vec{v}$$

Value of λ indicates 'variance' (spread) in direction of eigenvector v associated with λ

PCA: A Toy Example

- Consider a new co-ordinate system with one axis along the line
- All co-ordinates except the first one are zeros now.

3.7	7.5	11.2	15	18.7	22.4
0	0	0	0	0	0
0	0	0	0	0	0



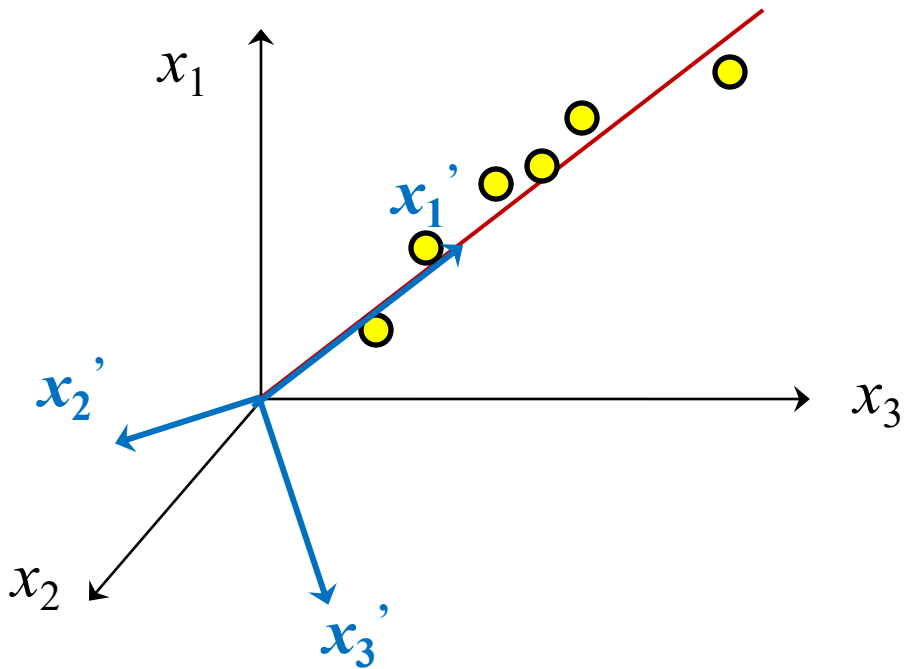
PCA: Toy Example - 2

1	4	3	5.7	5.1	2.2
2	7.9	5.8	12	9.9	4.1
3.1	12	9	18	15	6.3

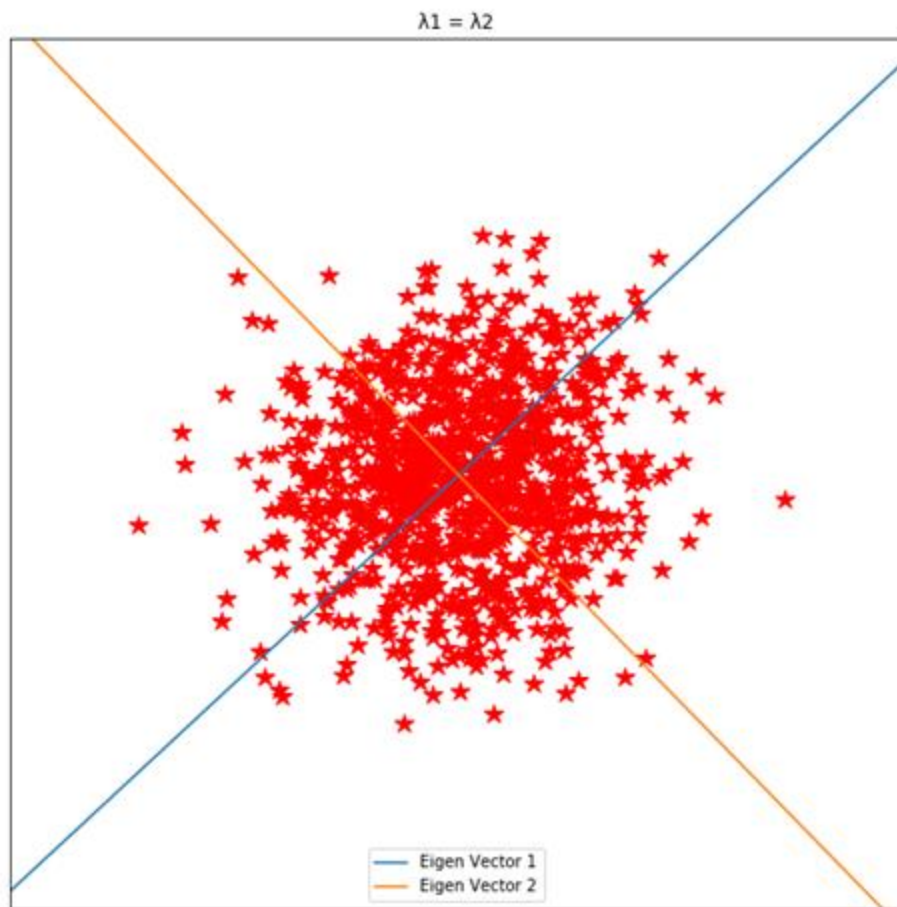


3.61	7.4	11.1	15.0	18.4	22.4
0.2	0.4	0.9	0.7	0.8	0.3
0.1	0.1	0.1	0.1	0.1	0.1

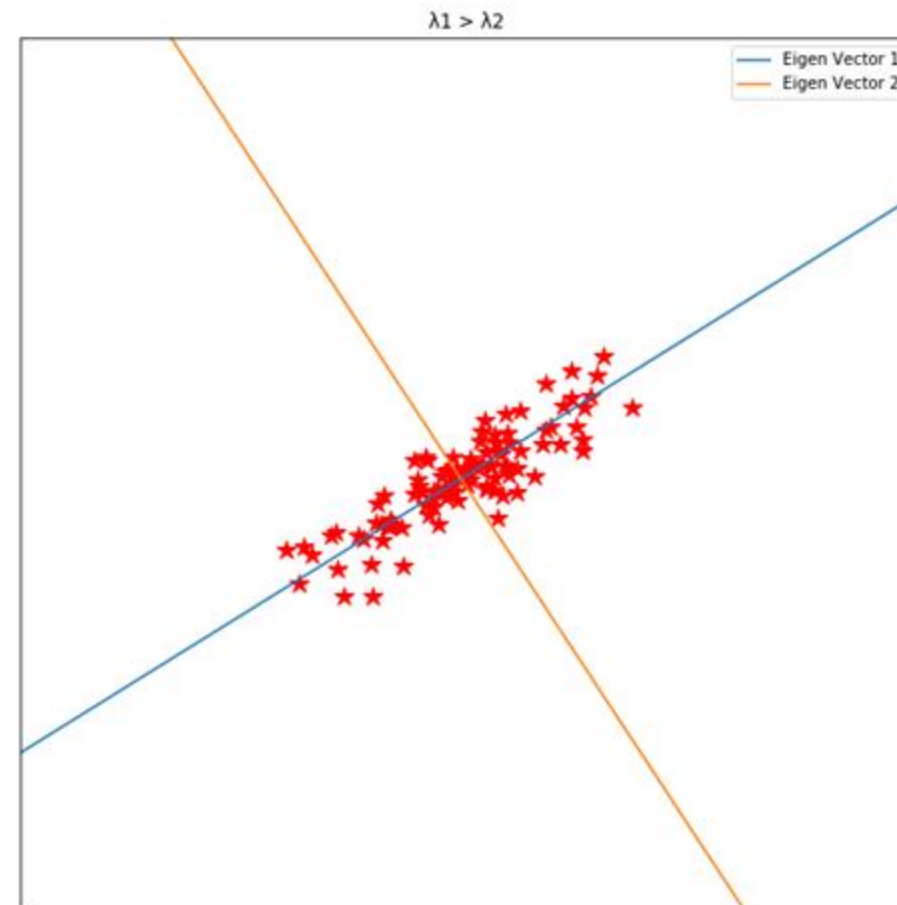
NOTE: These values are made up. Not exact.



Covariance, Eigen Values and Vectors

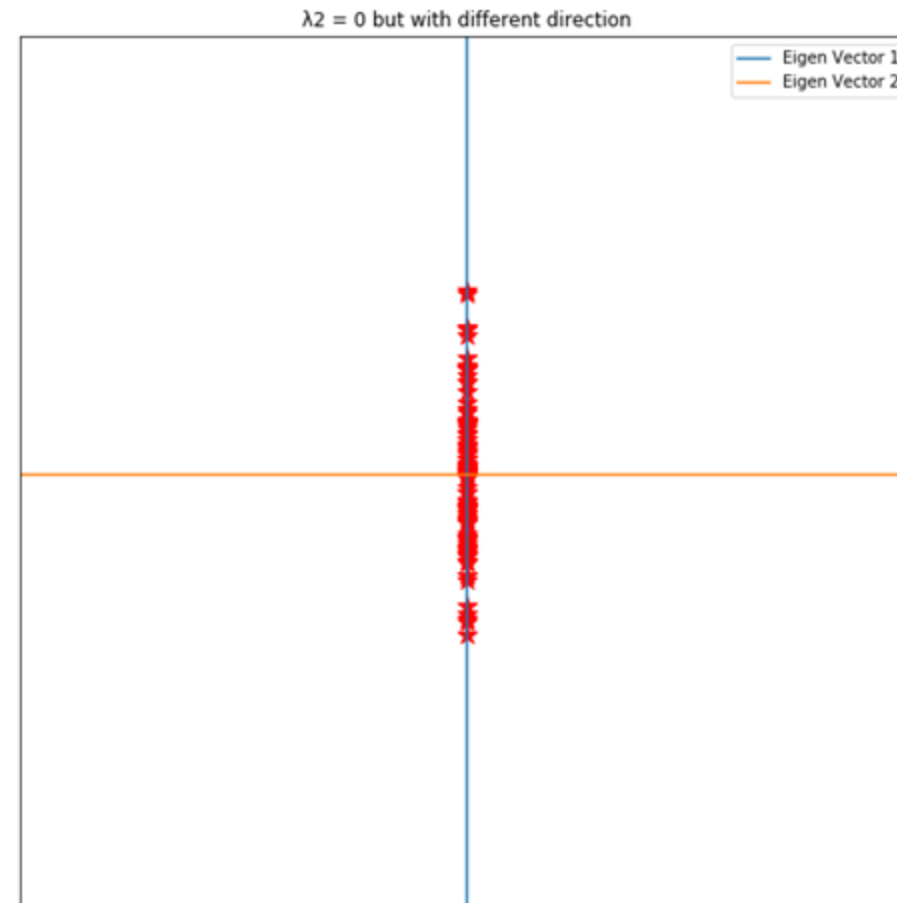
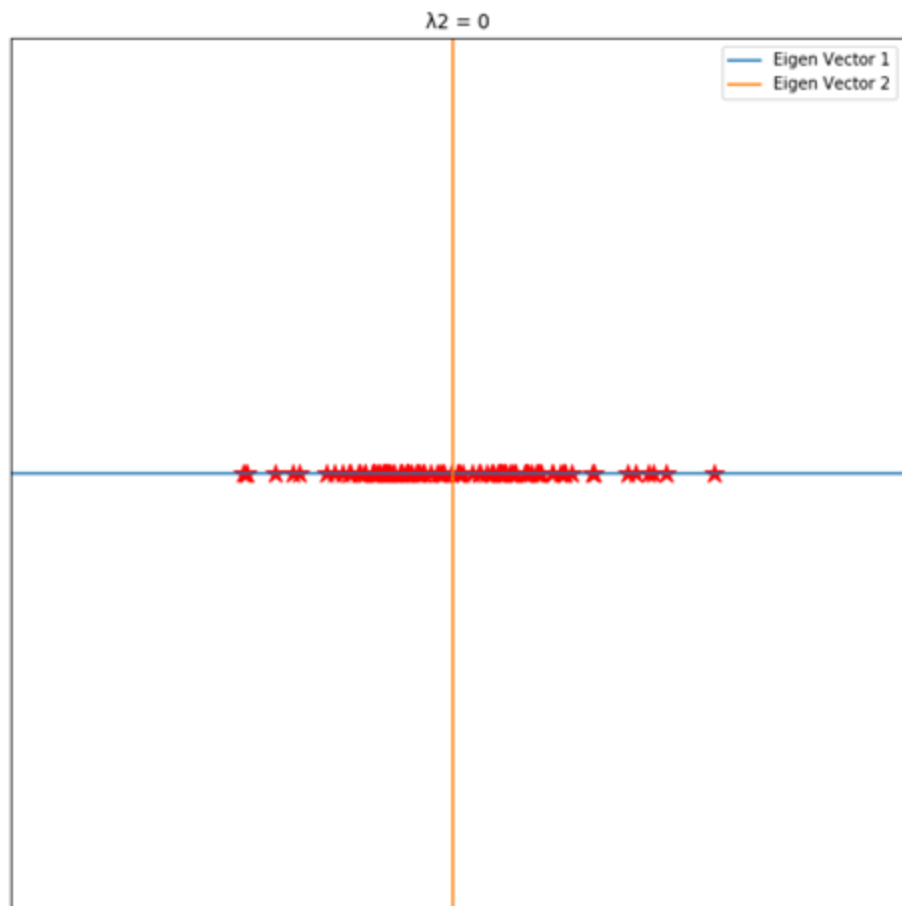


Both the Eigen values are equal
(Distribution is circular)



One Eigen value is greater than the other
(Distribution is elongated in the direction
of that Eigen vector)

Covariance, Eigen Values and Vectors



Only one Eigen value is non-zero,
distribution of data will align on that Eigen
vector

The PCA Recipe

1. Center the data



$$\begin{bmatrix} 0.39701 & 0.51117 \\ 0.55582 & 0.93003 \\ 0.59403 & 0.96645 \\ 0.51544 & 0.29759 \\ 0.85313 & 0.18118 \\ 0.88564 & 0.69114 \end{bmatrix}$$

$$\mathbf{X}' = \mathbf{X} - \bar{x}$$

2. Compute the covariance matrix

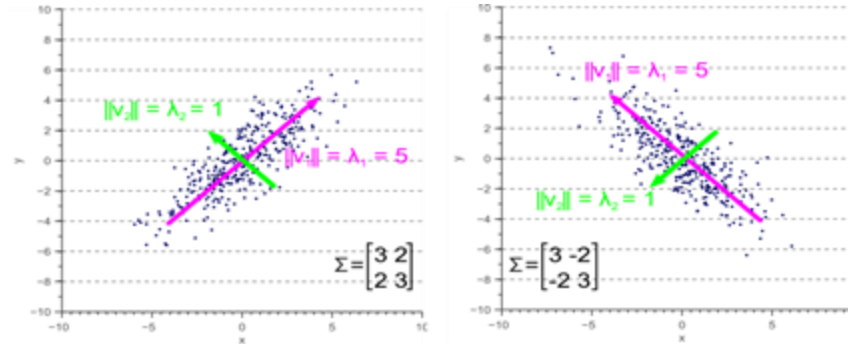
$$\begin{pmatrix} & M1 & M2 & M3 & \dots & Mn \\ S1 & q_{1,1} & q_{1,2} & q_{1,3} & \dots & q_{1,n} \\ S2 & q_{2,1} & q_{2,2} & q_{2,3} & \dots & q_{2,n} \\ S3 & q_{3,1} & q_{3,2} & q_{3,3} & \dots & q_{3,n} \\ \dots & \dots & \dots & \dots & \dots & \dots \\ Sm & q_{m,1} & q_{m,2} & q_{m,3} & \dots & q_{m,n} \end{pmatrix} \Rightarrow C = \begin{pmatrix} \text{cov}(M_1, M_1) & \text{cov}(M_1, M_2) & \dots & \text{cov}(M_1, M_n) \\ \text{cov}(M_2, M_1) & \text{cov}(M_2, M_2) & \dots & \text{cov}(M_2, M_n) \\ \vdots & \vdots & \ddots & \vdots \\ \text{cov}(M_n, M_1) & \text{cov}(M_n, M_2) & \dots & \text{cov}(M_n, M_n) \end{pmatrix}_{n \times n}$$

N-dimensional Covariance Matrix

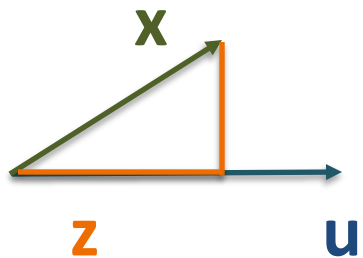
The PCA Recipe

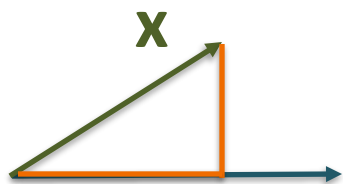
3. Compute Eigenvectors and Eigenvalues of Covariance Matrix Σ

$$\Sigma \vec{v} = \lambda \vec{v}$$



4. Project data onto eigenvectors to obtain new coordinates



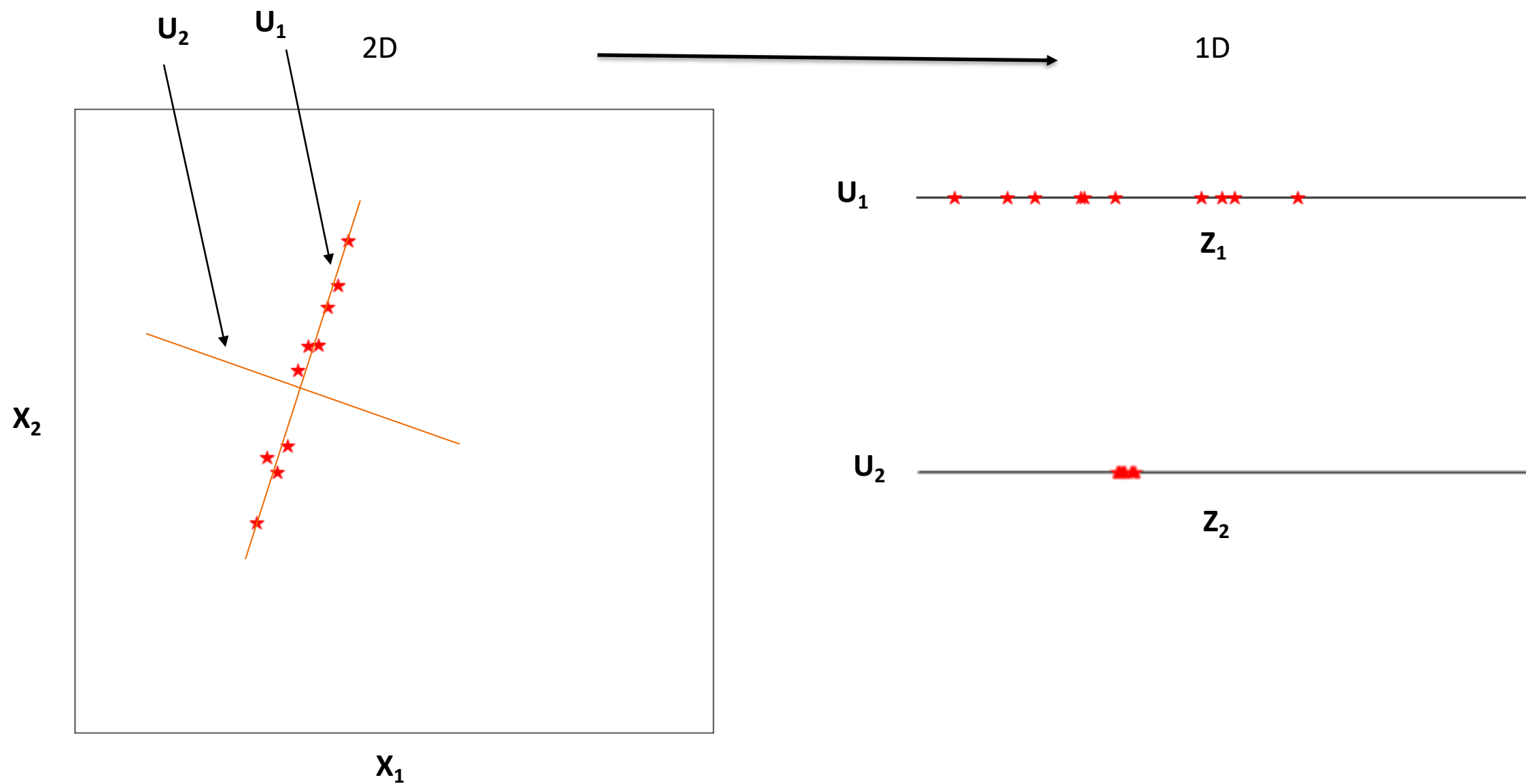


z **v (eigenvector)**

$$\begin{bmatrix} z_1 \\ z_2 \\ z_3 \\ z_4 \end{bmatrix} = \begin{bmatrix} \cdot & \cdot & v_1^T & \cdot & \cdot \\ \cdot & \cdot & v_2^T & \cdot & \cdot \\ \cdot & \cdot & v_3^T & \cdot & \cdot \\ \cdot & \cdot & v_4^T & \cdot & \cdot \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

New
coordinates

Old
coordinates



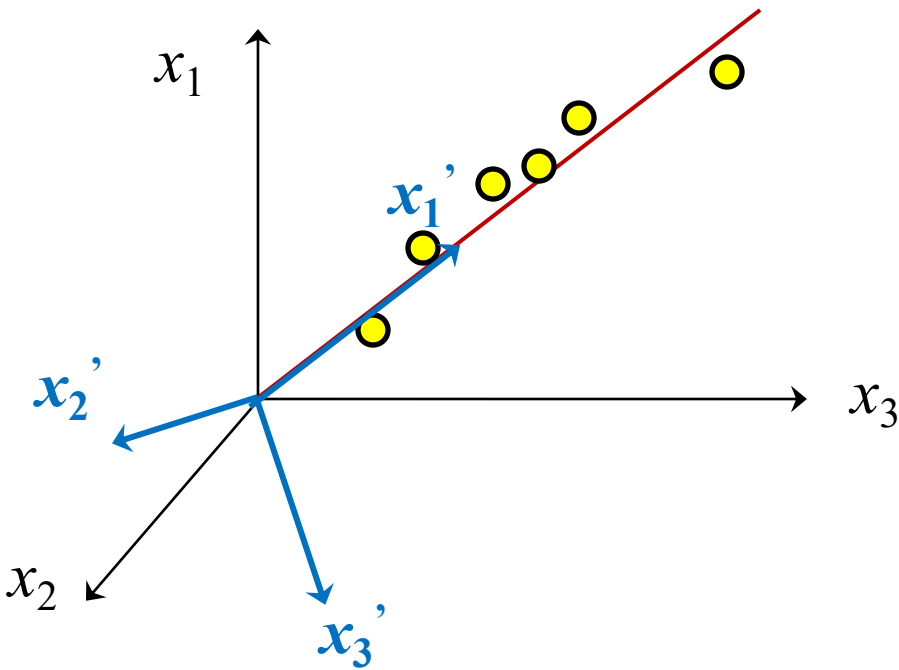
PCA: Toy Example - 2

1	4	3	5.7	5.1	2.2
2	7.9	5.8	12	9.9	4.1
3.1	12	9	18	15	6.3

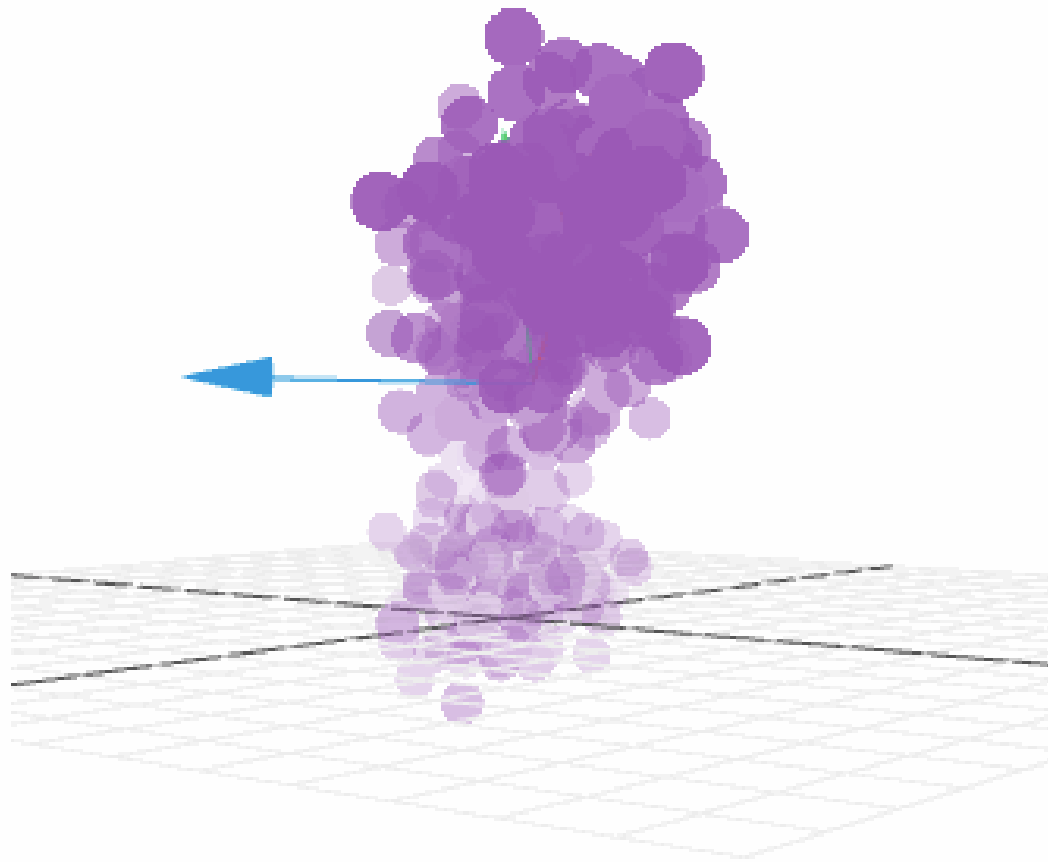


3.61	7.4	11.1	15.0	18.4	22.4
0.2	0.4	0.9	0.7	0.8	0.3
0.1	0.1	0.1	0.1	0.1	0.1

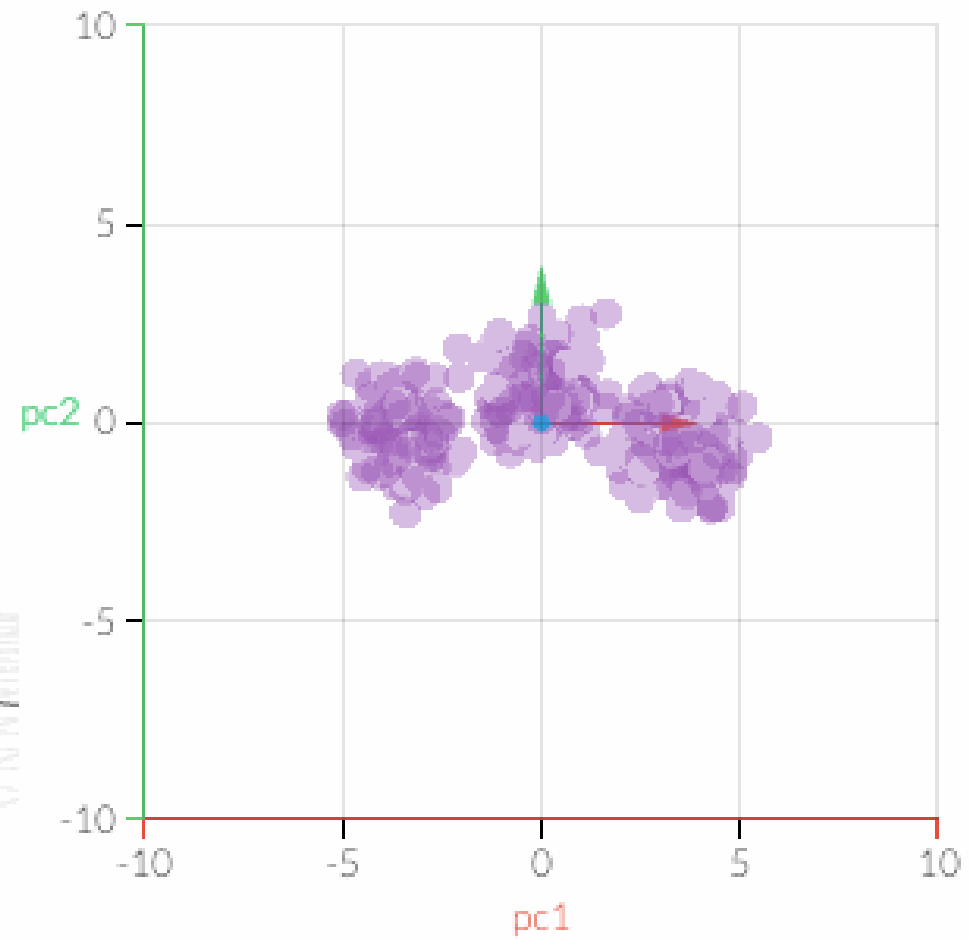
NOTE: These values are made up. Not exact.



3D to 2D



X_1, X_2, X_3



Z_1, Z_2

PCA based Feature Extraction

$r \times 1$

$r \times d$

$d \times 1$

$$\begin{matrix} z_1 \\ z_2 \\ z_3 \\ \vdots \\ \vdots \\ z_r \end{matrix} = \begin{matrix} u_{11} & u_{12} & u_{13} & \cdot & \cdot & \cdot & \cdot & \cdot & u_{1d} \\ u_{21} & u_{22} & u_{23} & \cdot & \cdot & \cdot & \cdot & \cdot & u_{2d} \\ u_{31} & u_{32} & u_{33} & \cdot & \cdot & \cdot & \cdot & \cdot & u_{3d} \\ \cdot & & & & & & & & \\ u_{r1} & u_{r2} & u_{r3} & \cdot & \cdot & \cdot & \cdot & \cdot & u_{rd} \end{matrix}$$

Z U

Each row in U is an eigen vector of co-variance Matrix

X

Appreciating PCA: Two Questions

$$\text{Eg. } \frac{\sum_{i=1}^r \lambda_i}{\sum_{i=1}^d \lambda_i} > 0.90$$

- How many Eigen vectors to select?
 - Ans: Eigen Vectors corresponding to the larger Eigen values
- How much information is lost? Can we recover the old data/information from the new?

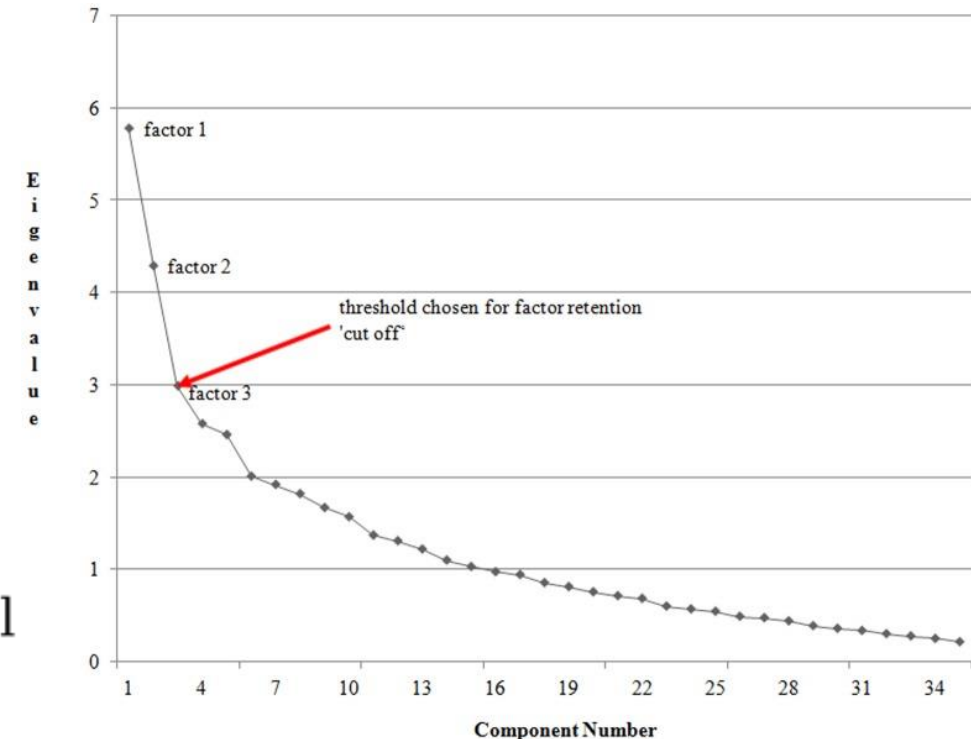
$$\mathbf{x} = z_1 \mathbf{u}_1 + z_2 \mathbf{u}_2 + z_3 \mathbf{u}_3 + z_4 \mathbf{u}_4$$

$$\mathbf{x} = z_1 \mathbf{u}_1 + z_2 \mathbf{u}_2 + z_3 \mathbf{u}_3 + z_4 \mathbf{u}_4$$

$$\mathbf{x}' = z_1 \mathbf{u}_1 + z_2 \mathbf{u}_2$$

$$\text{Loss in Information} = \|\mathbf{x} - \mathbf{x}'\|$$

Note: z_3 and z_4 are small and also λ_3 and λ_4 are small



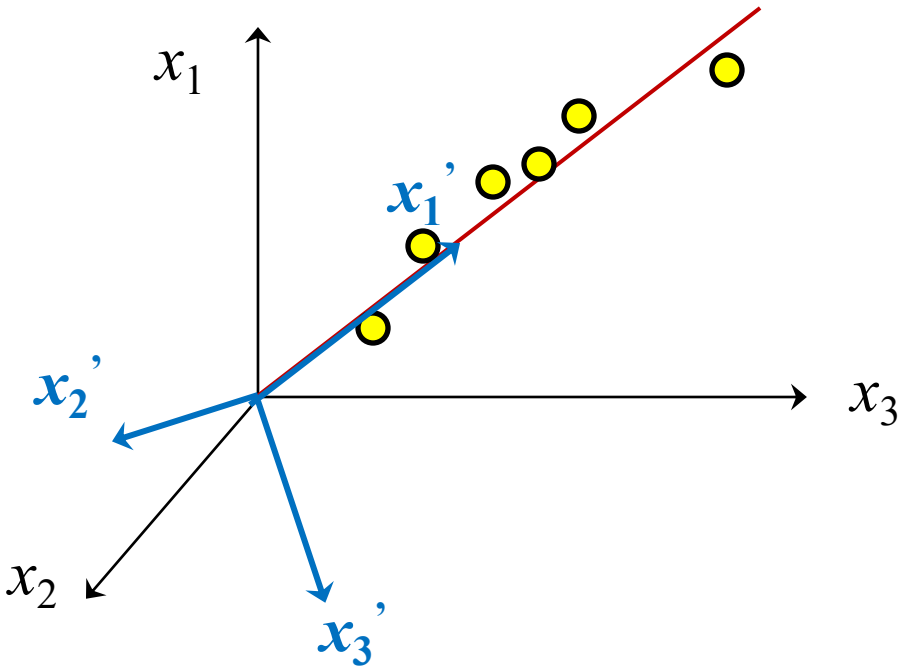
PCA: Toy Example - 2

1	4	3	5.7	5.1	2.2
2	7.9	5.8	12	9.9	4.1
3.1	12	9	18	15	6.3



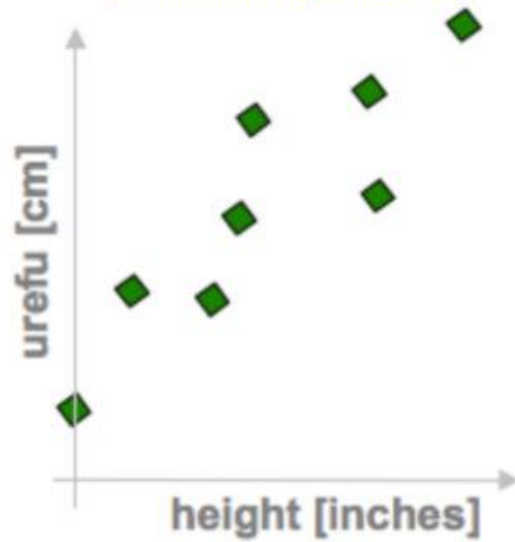
3.61	7.4	11.1	15.0	18.4	22.4
0.2	0.4	0.9	0.7	0.8	0.3
0.1	0.1	0.1	0.1	0.1	0.1

NOTE: These values are made up. Not exact.

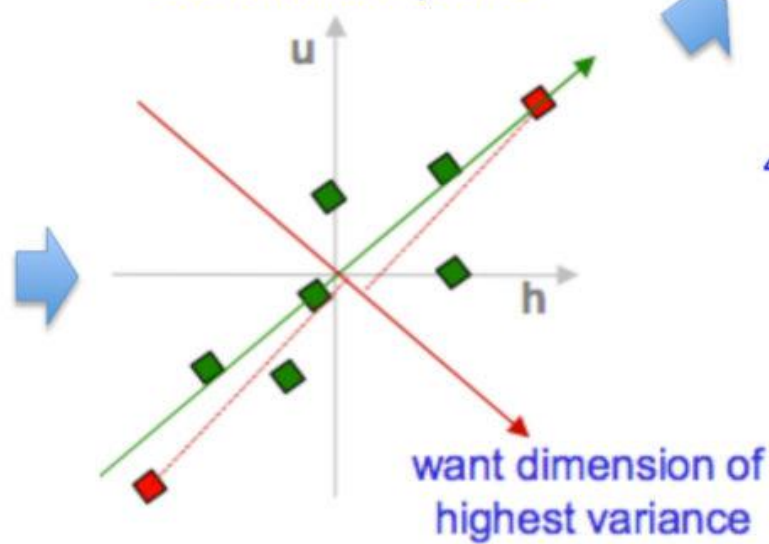


PCA in a nutshell

1. correlated hi-d data
(“urefu” means “height” in Swahili)



2. center the points



3. compute covariance matrix

$$\begin{matrix} & h & u \\ h & 2.0 & 0.8 \\ u & 0.8 & 0.6 \end{matrix} \rightarrow \text{cov}(h, u) = \frac{1}{n} \sum_{i=1}^n h_i u_i$$

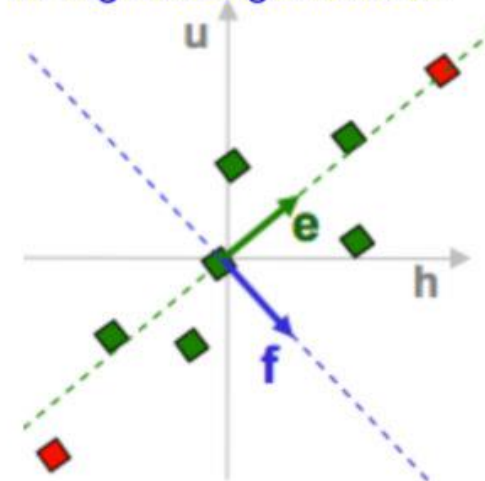
4. eigenvectors + eigenvalues

$$\begin{pmatrix} 2.0 & 0.8 \\ 0.8 & 0.6 \end{pmatrix} \begin{pmatrix} e_h \\ e_u \end{pmatrix} = \lambda_e \begin{pmatrix} e_h \\ e_u \end{pmatrix}$$

$$\begin{pmatrix} 2.0 & 0.8 \\ 0.8 & 0.6 \end{pmatrix} \begin{pmatrix} f_h \\ f_u \end{pmatrix} = \lambda_f \begin{pmatrix} f_h \\ f_u \end{pmatrix}$$

`eig(cov(data))`

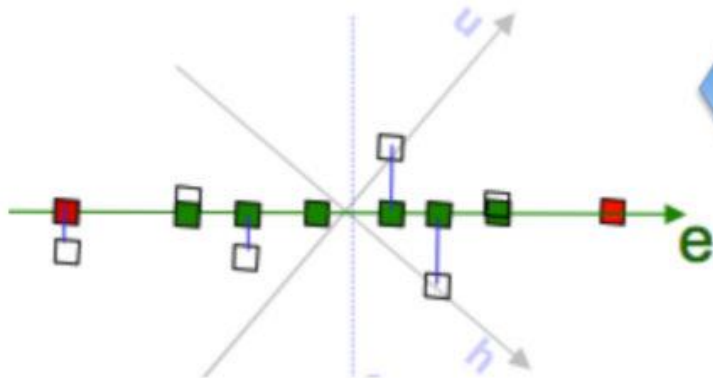
5. pick $m < d$ eigenvectors
w. highest eigenvalues



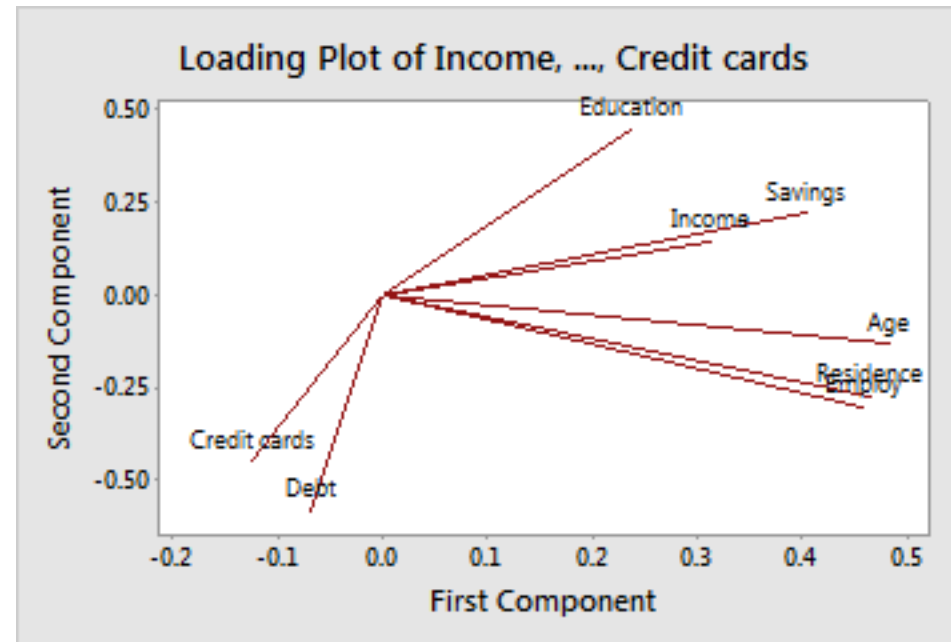
6. project data points to
those eigenvectors

$$x'_e = x^T e = \sum_{j=1}^d x_{ij} e_j$$

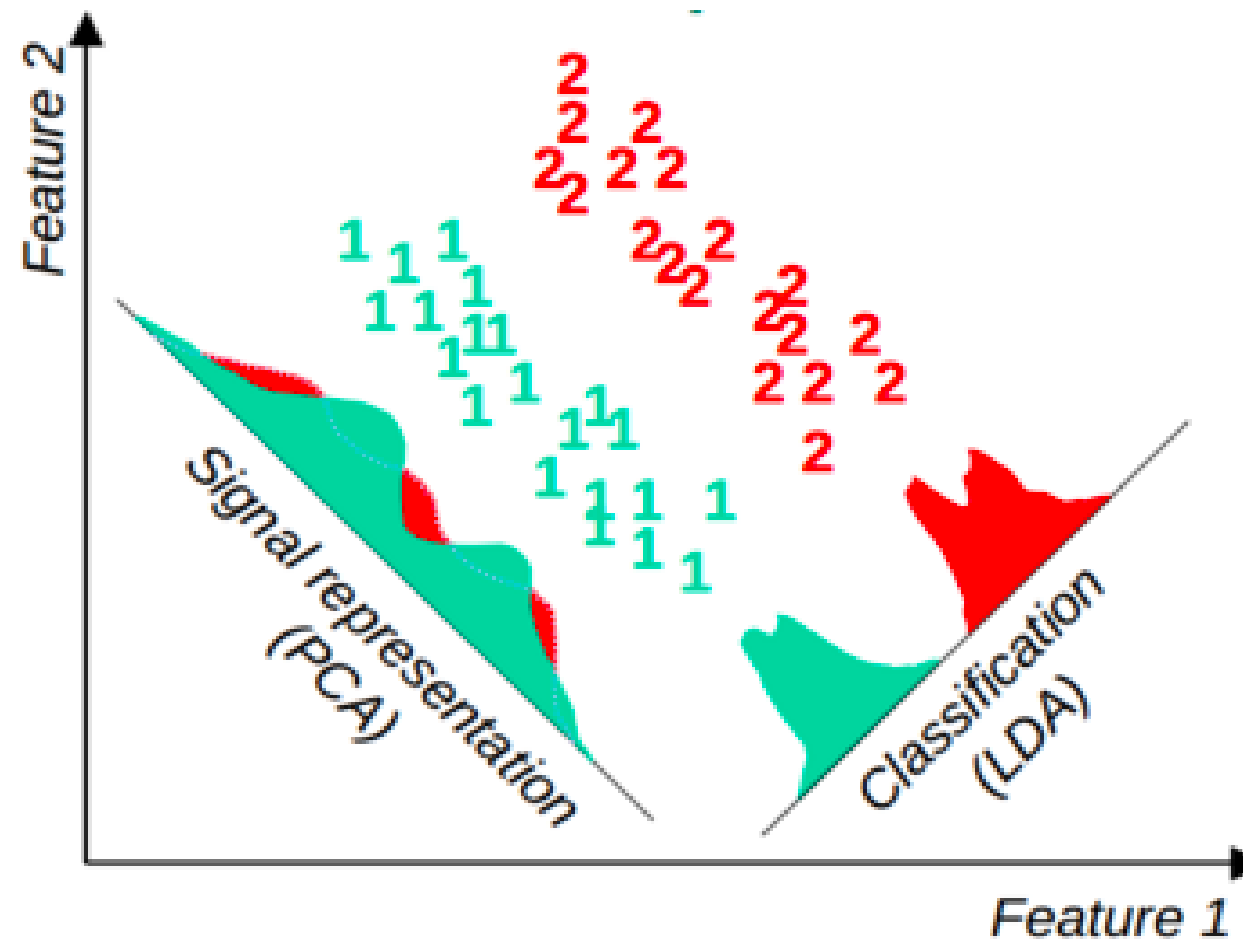
7. uncorrelated low-d data



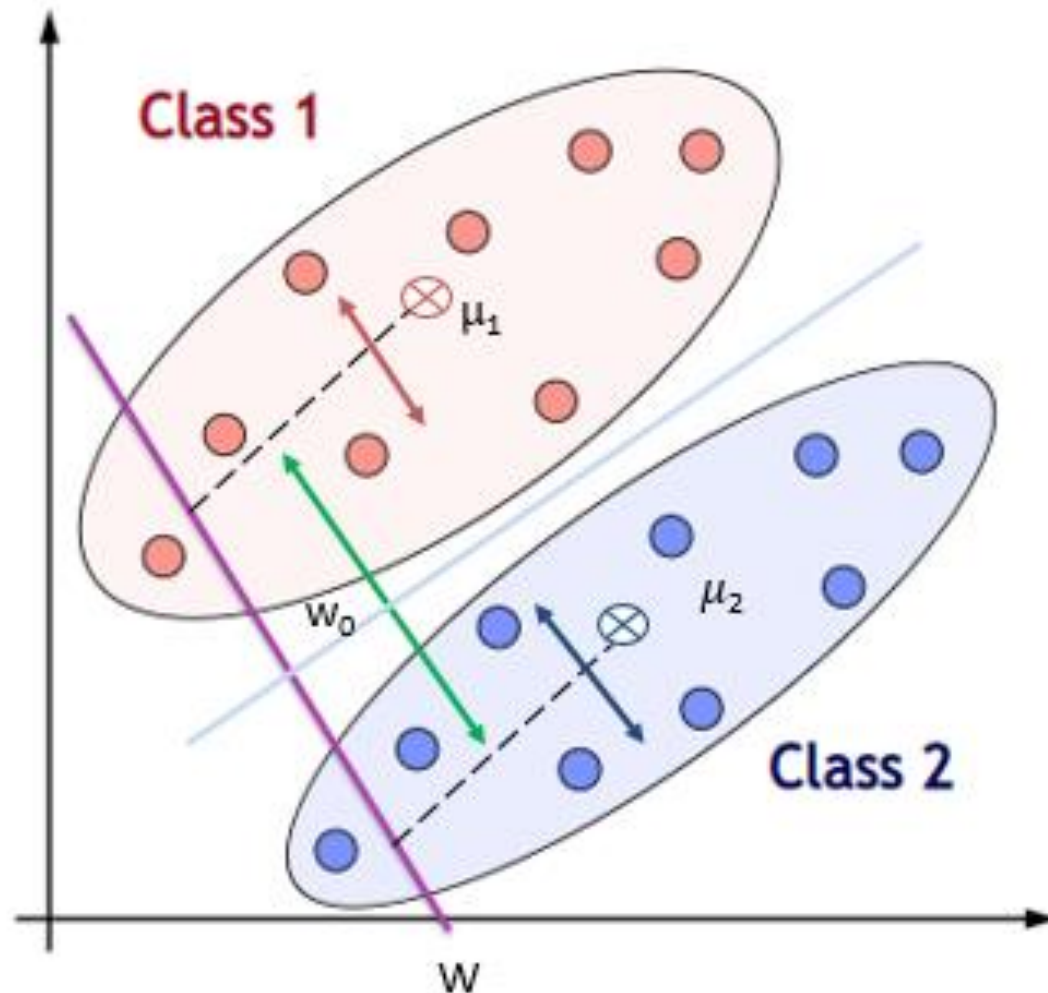
Analysis using 'factor loadings'



Linear Discriminant Analysis (LDA)



Linear Discriminant Analysis (LDA)



- Maximize distance between classes
- Minimize distance within a class
- Criterion: $J(w) = \frac{w^T S_b w}{w^T S_w w}$

S_b = between-class scatter matrix

S_w = within-class scatter matrix

- Vector w is a solution of generalized Eigen value problem:

$$S_b w = \lambda S_w w$$

- Classification function:

$$g(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + w_0 \begin{cases} \text{Class 1} \\ \geq 0 \\ \text{Class 2} \end{cases}$$

Linear Discriminant Analysis (LDA)

- Also known as “Fisher Discriminant”
- Does dimensionality Reduction
 - Also use the label “y”
 - Or Supervised Dimensionality Reduction
- There are also nonlinear Dimensionality Reduction schemes

Summary

- We often get raw data/logs/measurements
- Two problems:
 - Select good ones out of all
 - Define new ones as linear combination of existing
- Dimensionality reduction for
 - Compression/compaction
 - Classification/Discrimination

Selecting and Extracting Features

$$\begin{bmatrix} x_1 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

Selecting first and third feature

$$\begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = \begin{bmatrix} 0.1 & 0.2 & 0.3 & 0.4 \\ 0.0 & 0.4 & 0.2 & 1.7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

New Features as linear combination of old Features

$$X' = AX$$

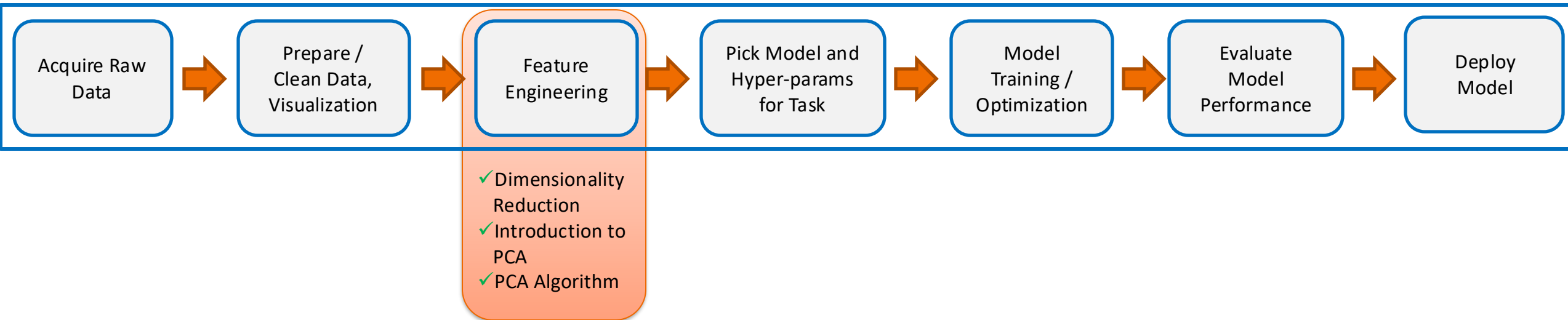
For PCA: Rows are Eigen vectors of the covariance matrix.

$$\begin{bmatrix} x_1 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

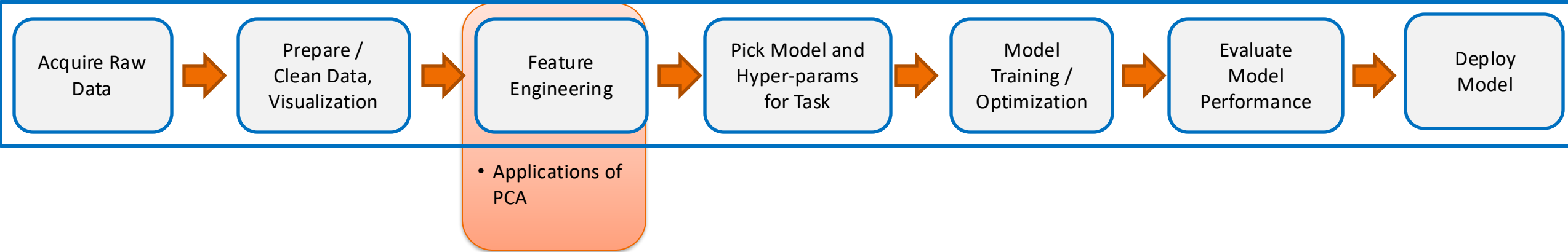
Selecting first and fourth feature

$$\begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = \begin{bmatrix} \cdots & u_1^T & \cdots \\ \cdots & u_2^T & \cdots \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

Journey so far...



Questions?



Dimensionality Reduction

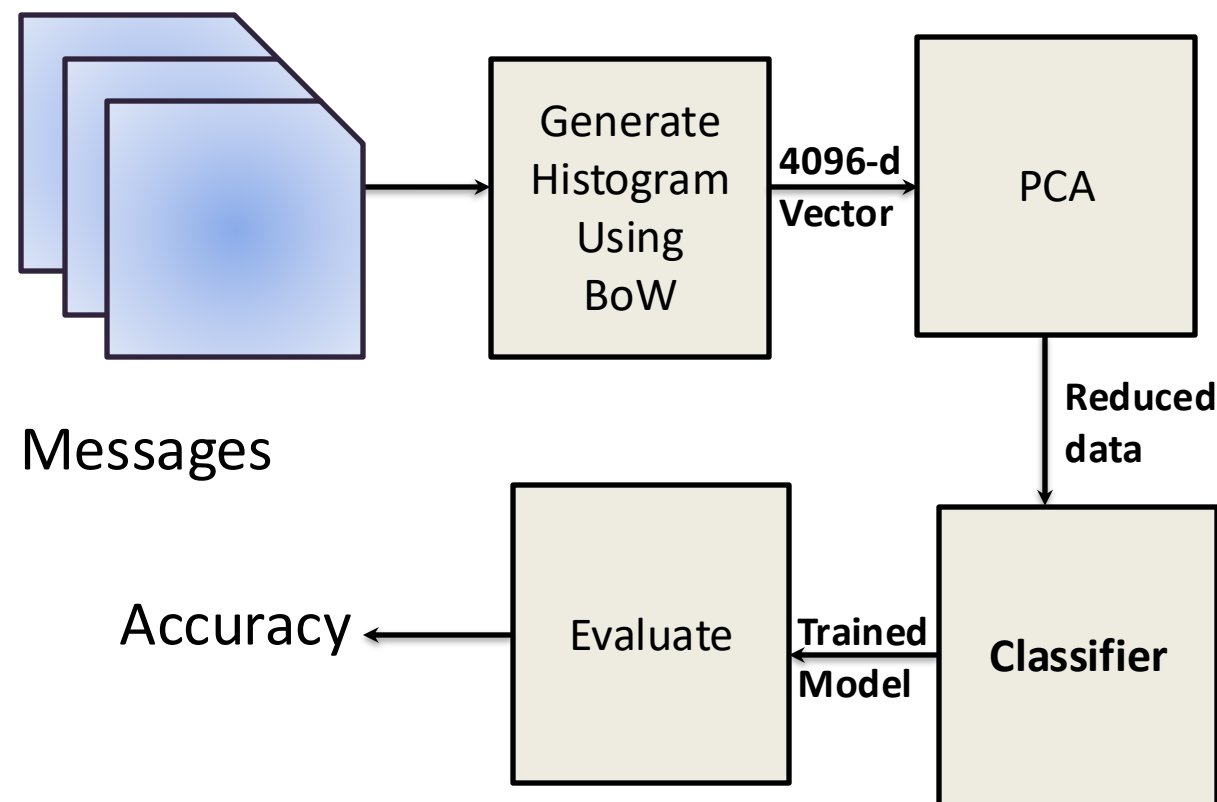
Applications of PCA

Case Study

Classification after PCA

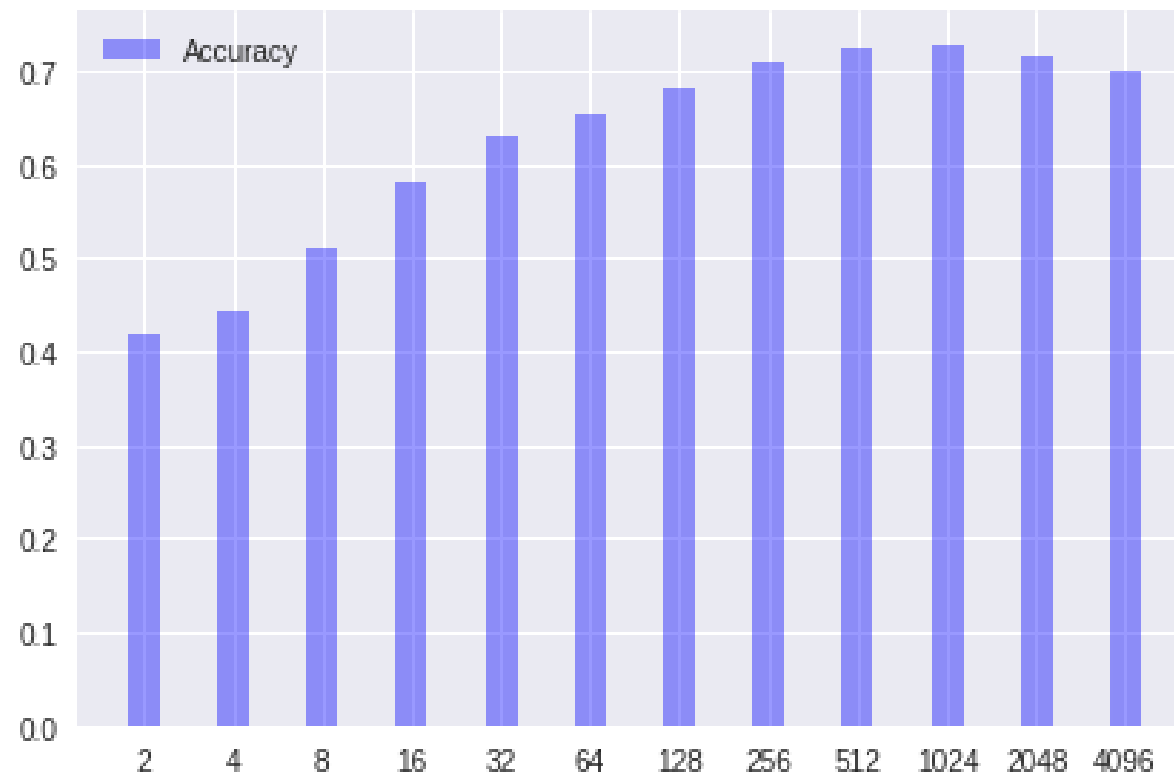
Case Study: PCA and Classification

- Text data with 20 classes
- Preprocessing:
 - Find the Histograms for Each Document using Bag of words
 - Apply PCA to reduce the dimensions
- Train the classifier on the reduced data
- Find the Accuracy to Evaluate the model



Effect of PCA on the Accuracy

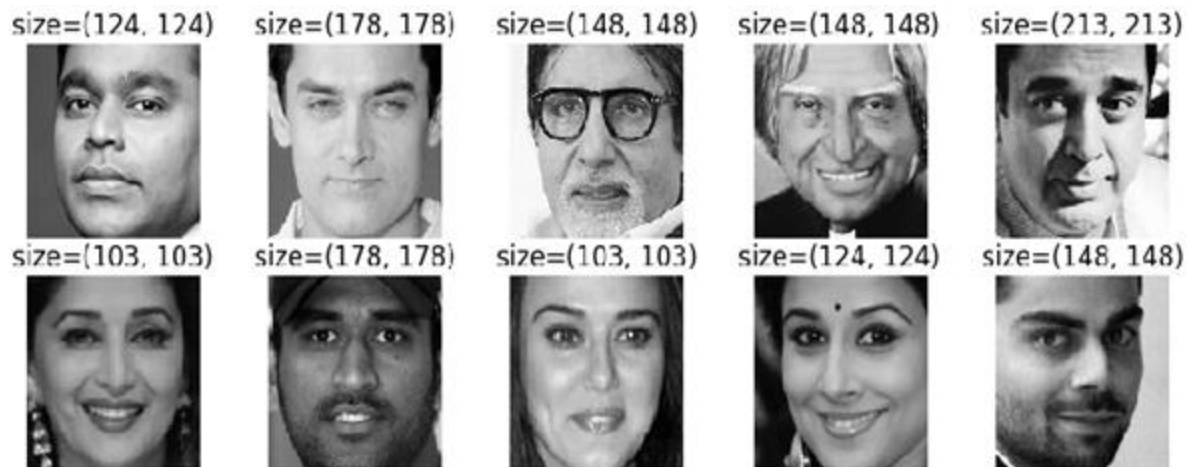
- Change r (dimensions in projected space) to 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024, 2048, 4096
- With just 3% (32) of the total dimensions (4096), comparable accuracies are obtained



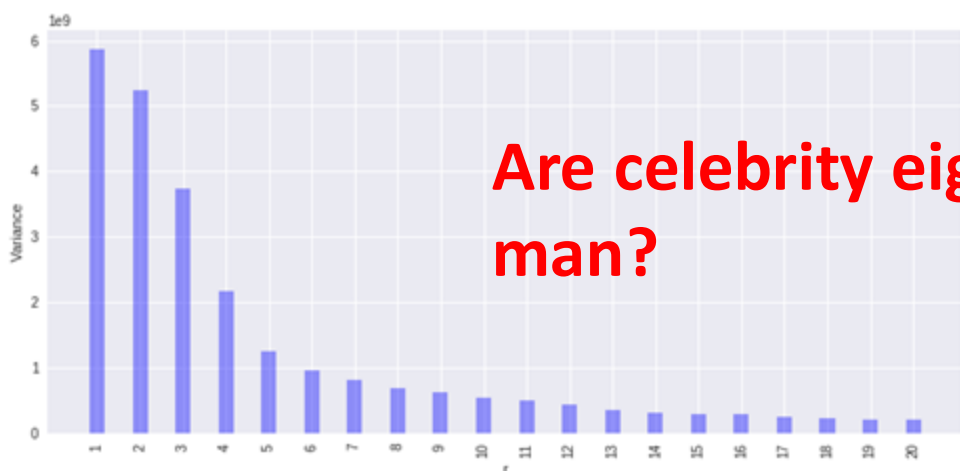
Case Study

Recognition and data compression with
Eigen faces

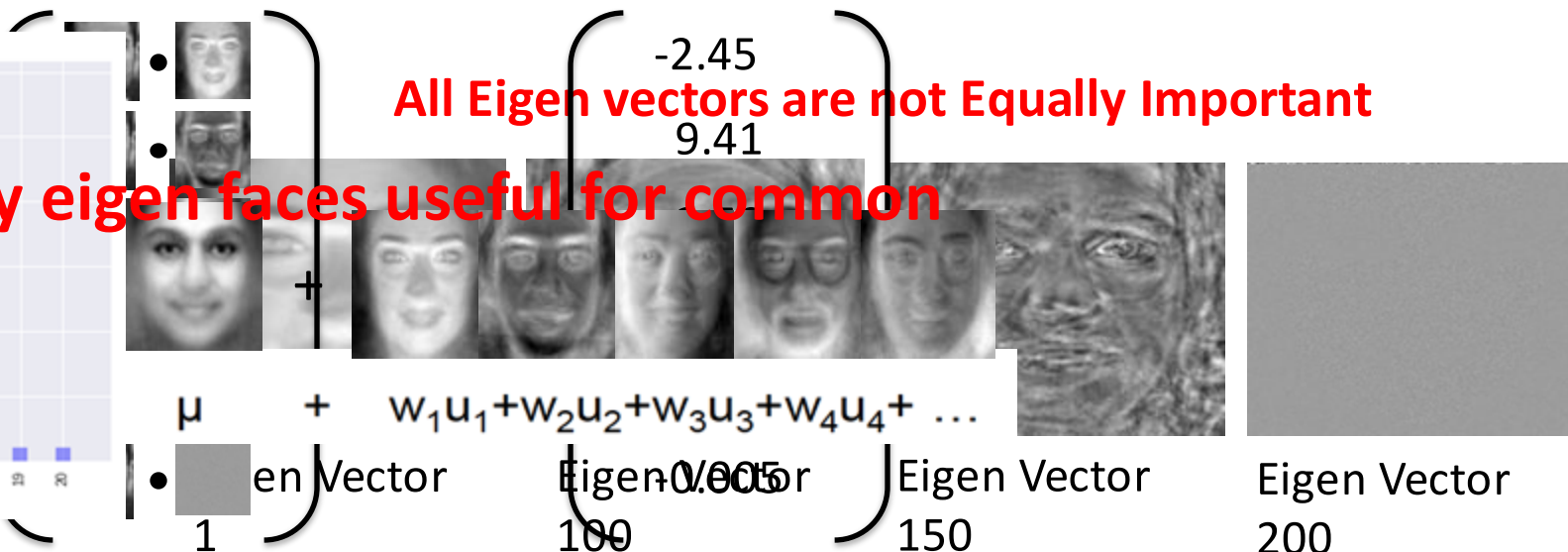
Recognize Indian Celebrities



- Rescale and Find mean of the face
- Find the Eigen Faces
 - All Eigen faces are not equally important
- Find the weights to represent the face in Eigen space
- Reconstruct the image



Are celebrity eigen faces useful for common man?

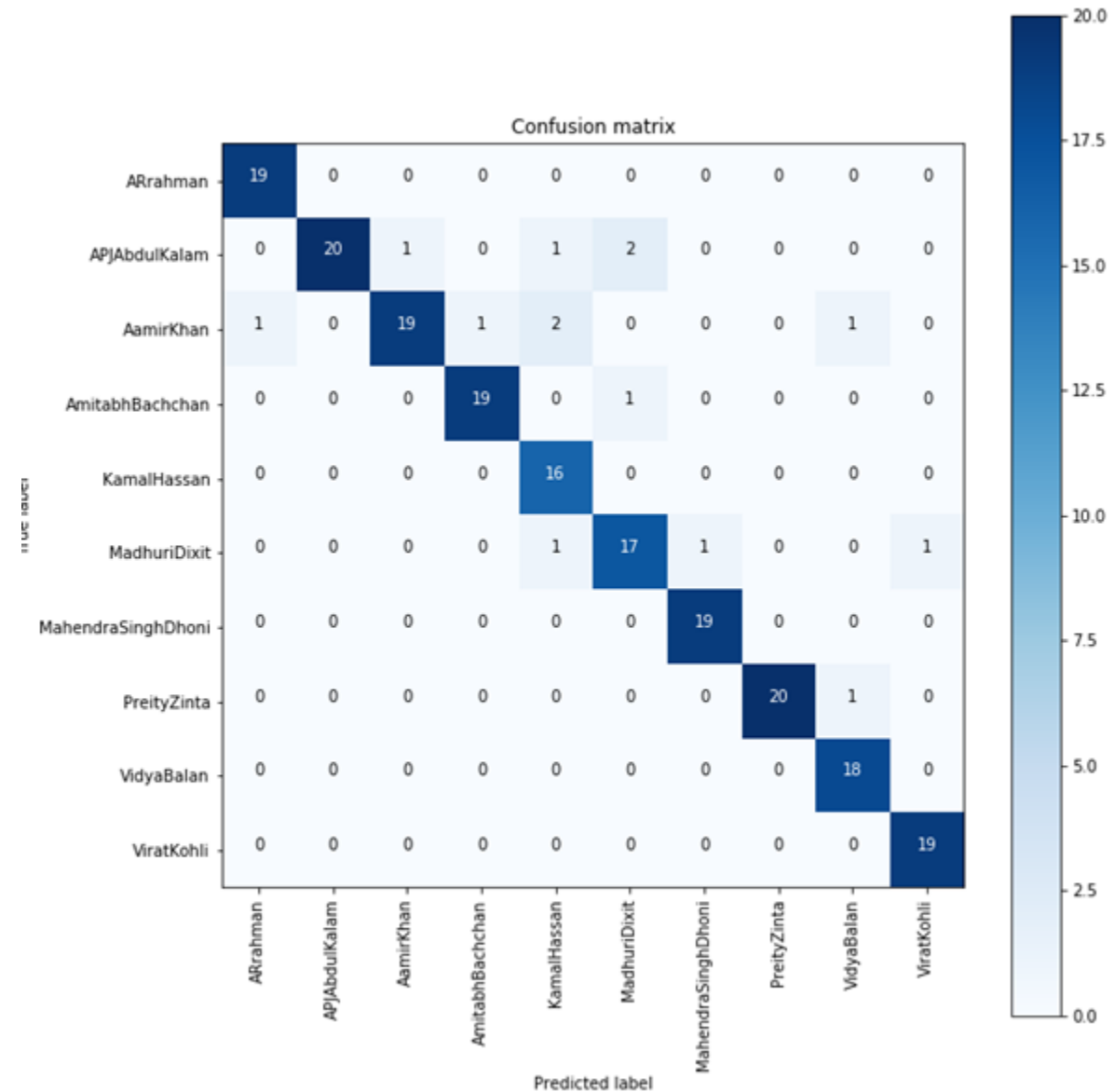


Classification

Training images: 400

Test images: 200

Accuracy: 96%



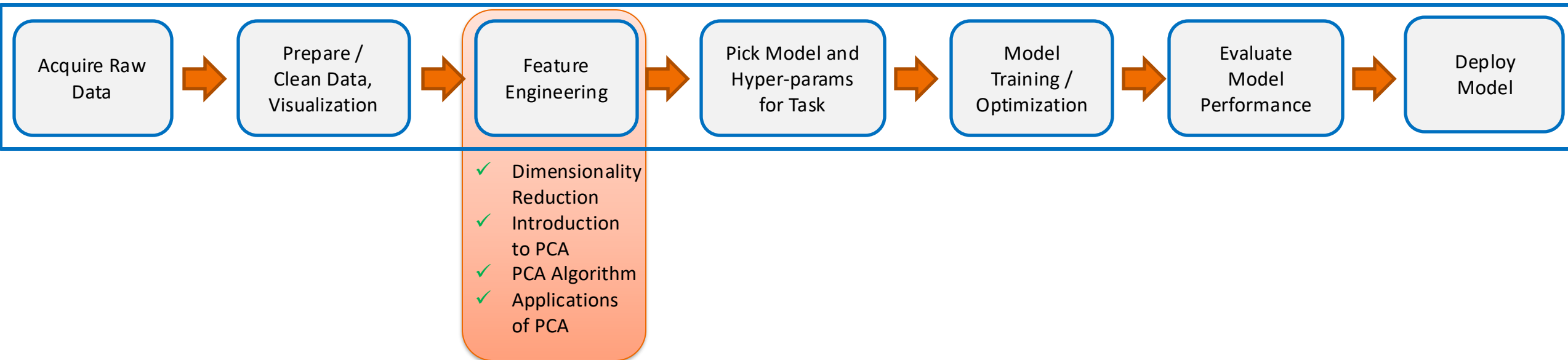
Three Viewpoints

- Maximal variance on the new features.
- Data Compression and Minimal Reconstruction Error.
- Orthogonal Line Fitting.

Summary: PCA

- Compute Eigen values and Eigen Vectors of the covariance matrix
- Select the principal components
- Define new features.
- Will classification performance improve? Depends:
 - Do we throw away signal?
 - Do we throw away noise?

Summary



Thanks!!

Questions?